Nebico Oil Retention Project (4.2)

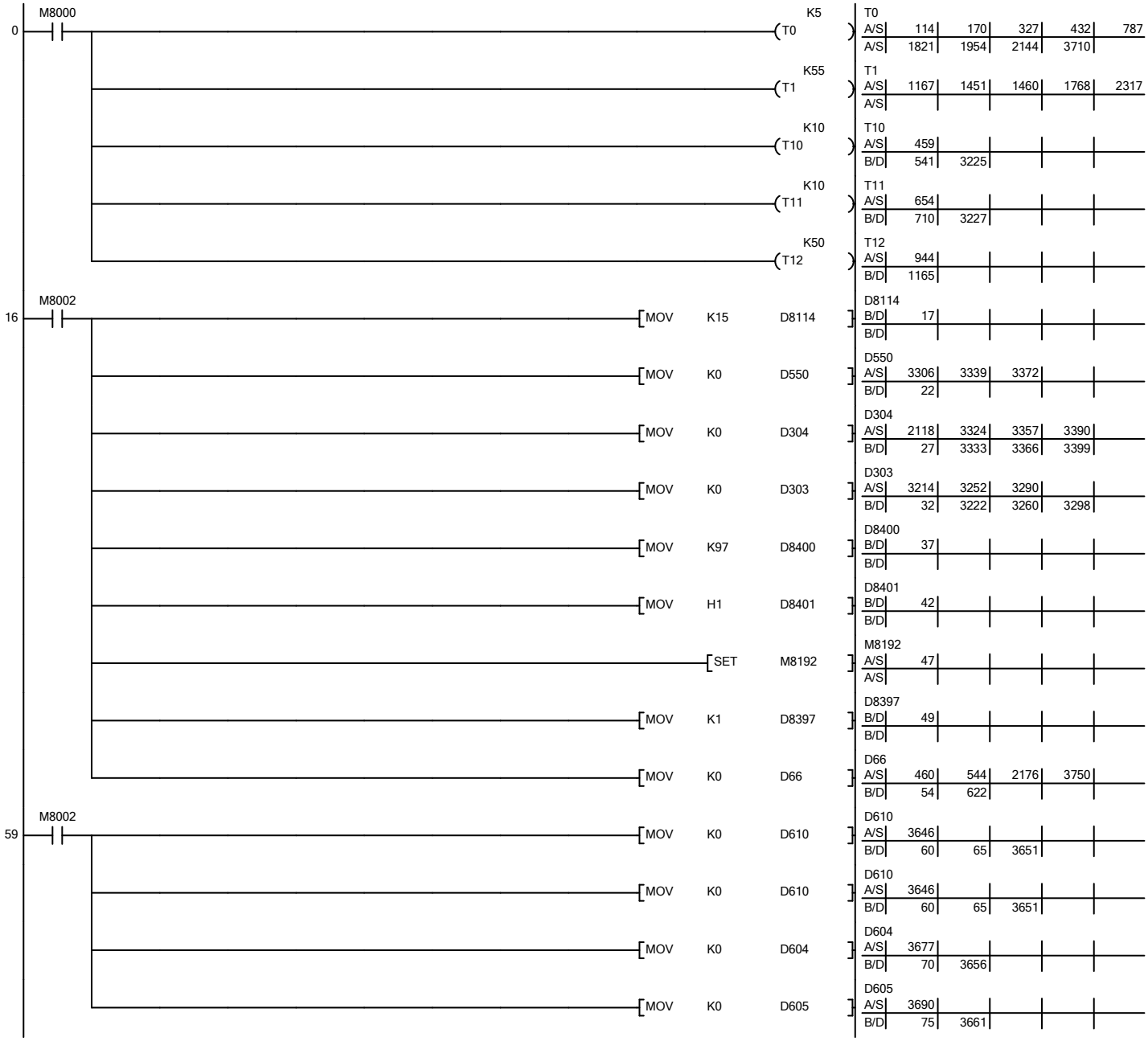
Coolmay L10S Ladder program
TK8050 x 3, TK8100 x 1

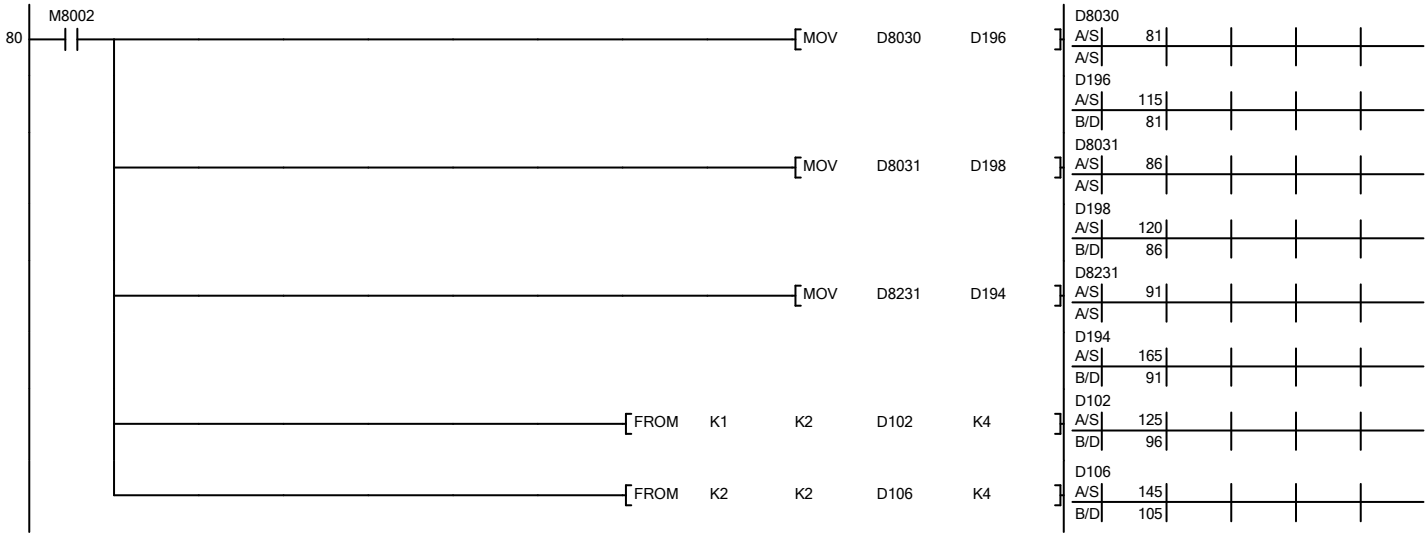
Modification:

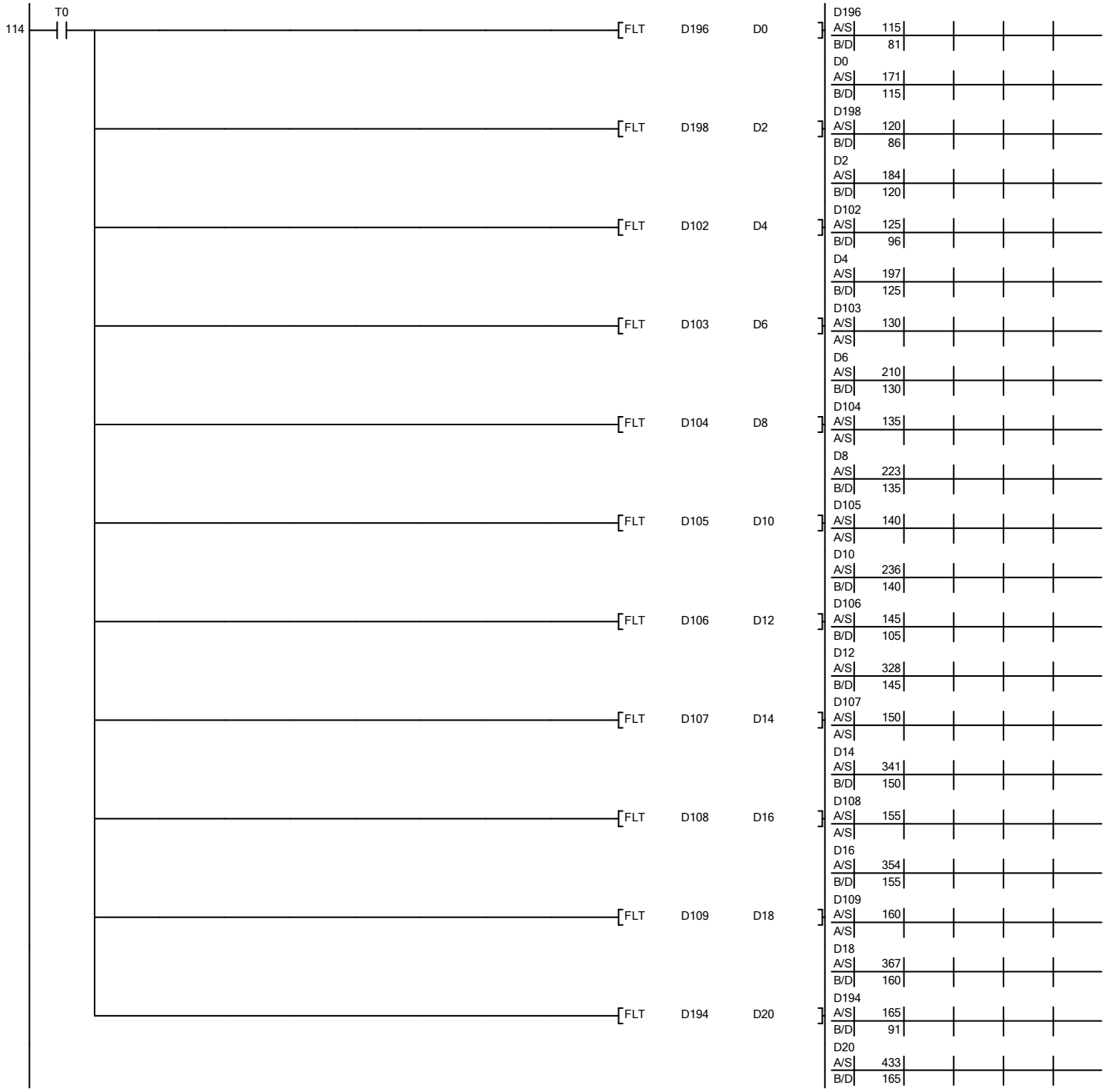
1. Removal of Z-Pointers
2. Max utilisation of General purpose data register
3. Corrected for Solenoid Valve change

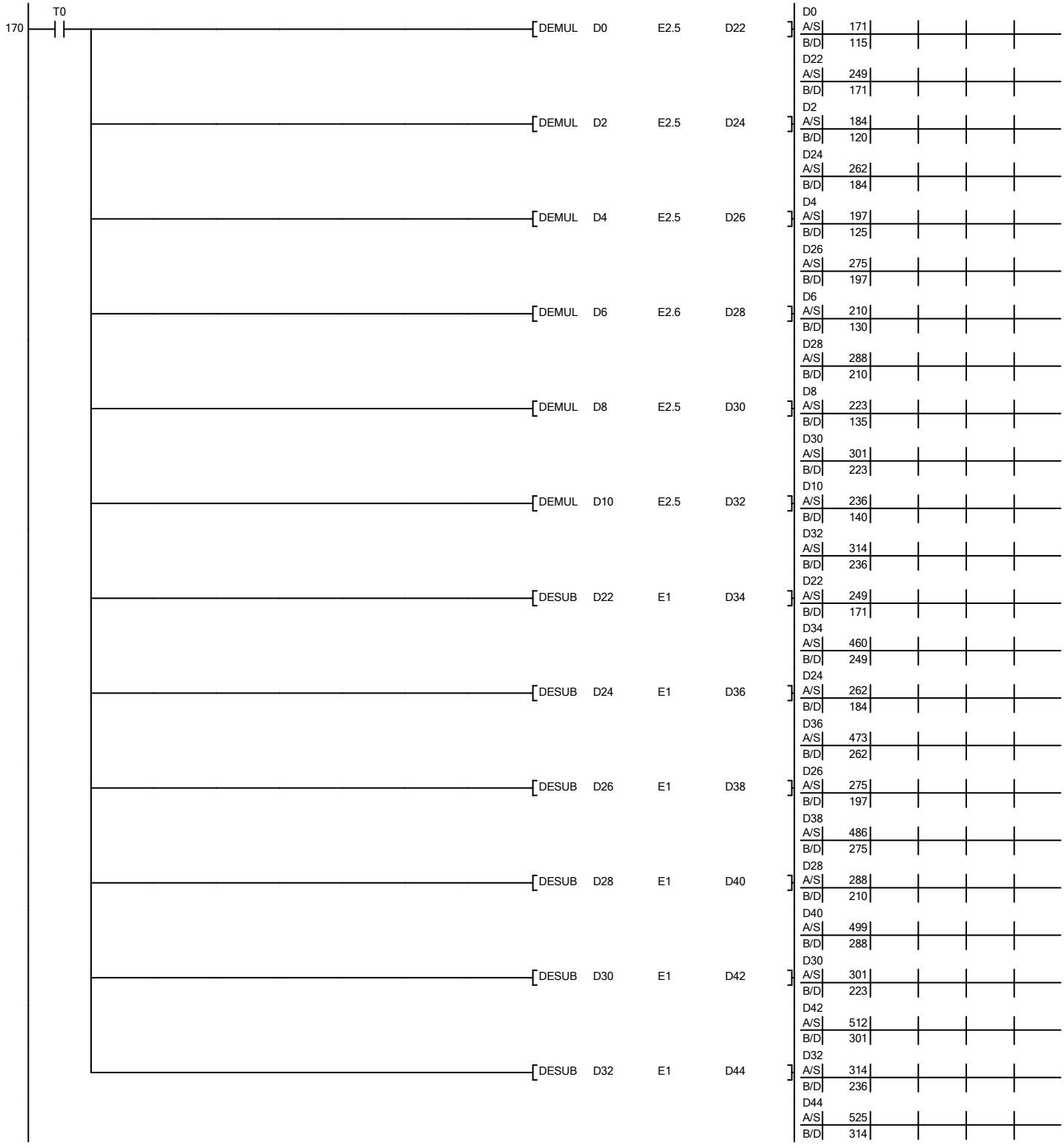
Prepared By: Sambridda Ranabhat (Architect & Programmer)

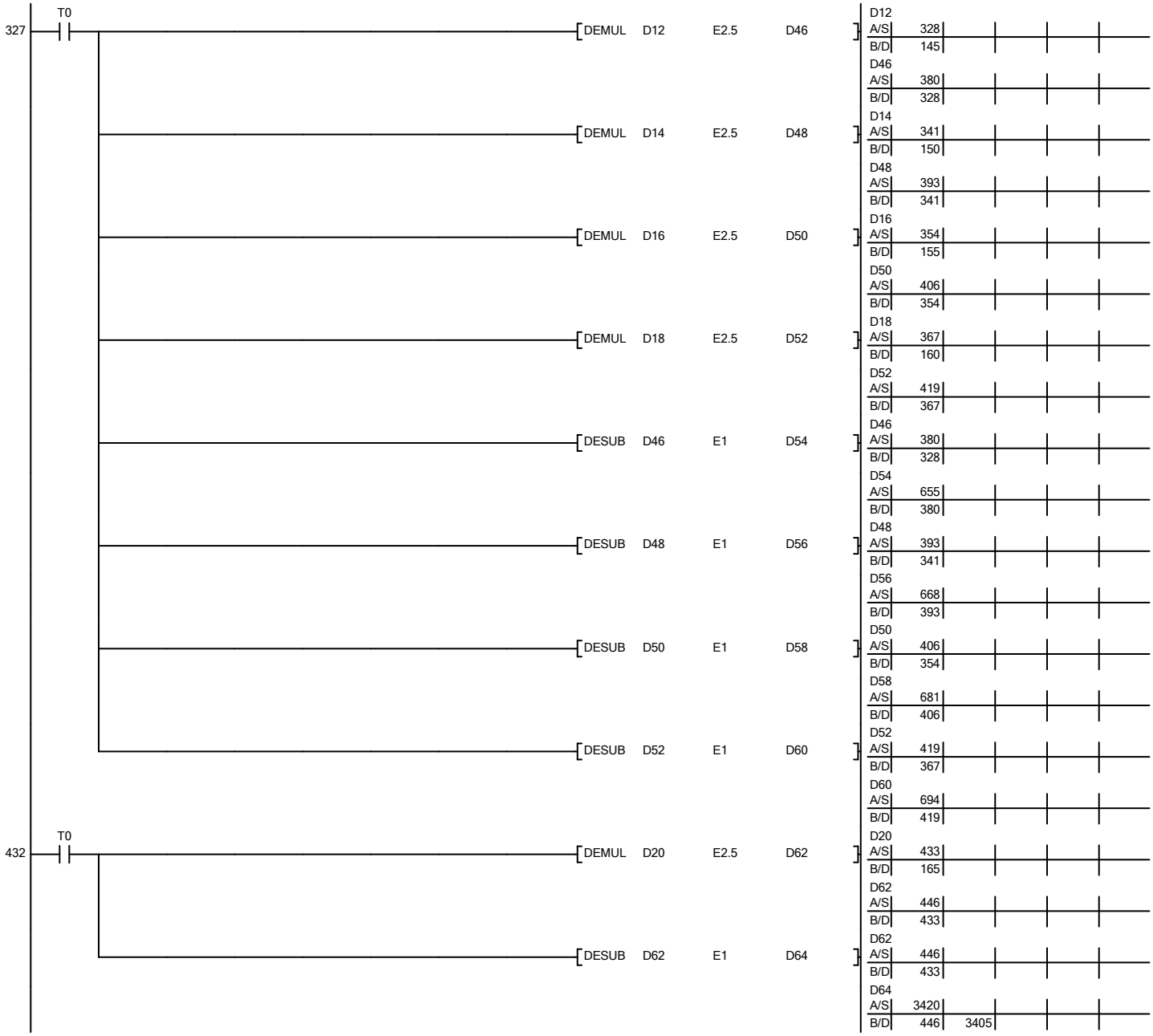
20/09/2026
Mechatronics Engineering .pvt .ltd

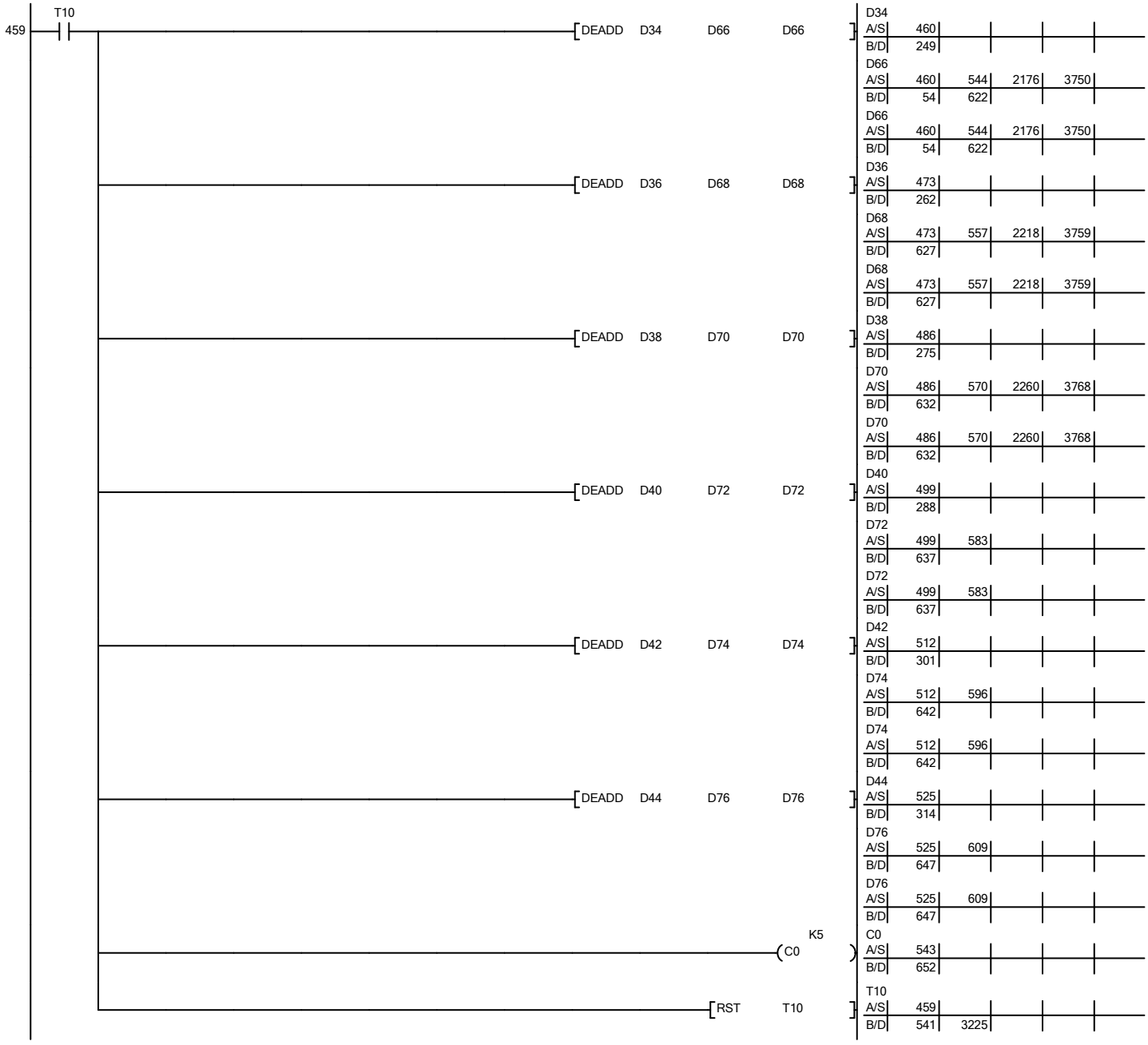

0
M8000
T0
K5
T1
K55
T10
K10
T11
K10
T12
K50
16
M8002
MOV K15 D8114
MOV K0 D550
MOV K0 D304
MOV K0 D303
MOV K97 D8400
MOV H1 D8401
SET M8192
MOV K1 D8397
MOV K0 D66
59
M8002
MOV K0 D610
MOV K0 D610
MOV K0 D604
MOV K0 D605
T0 A/S 114 170 327 432 787 A/S 1821 1954 2144 3710
T1 A/S 1167 1451 1460 1768 2317 A/S
T10 A/S 459 B/D 541 3225
T11 A/S 654 B/D 710 3227
T12 A/S 944 B/D 1165
D8114 B/D 17 B/D
D550 A/S 3306 3339 3372 B/D 22
D304 A/S 2118 3324 3357 3390 B/D 27 3333 3366 3399
D303 A/S 3214 3252 3290 B/D 32 3222 3260 3298
D8400 B/D 37 B/D
D8401 B/D 42 B/D
M8192 A/S 47 A/S
D8397 B/D 49 B/D
D66 A/S 460 544 2176 3750 B/D 54 622
D610 A/S 3646 B/D 60 65 3651
D610 A/S 3646 B/D 60 65 3651
D604 A/S 3677 B/D 70 3656
D605 A/S 3690 B/D 75 3661

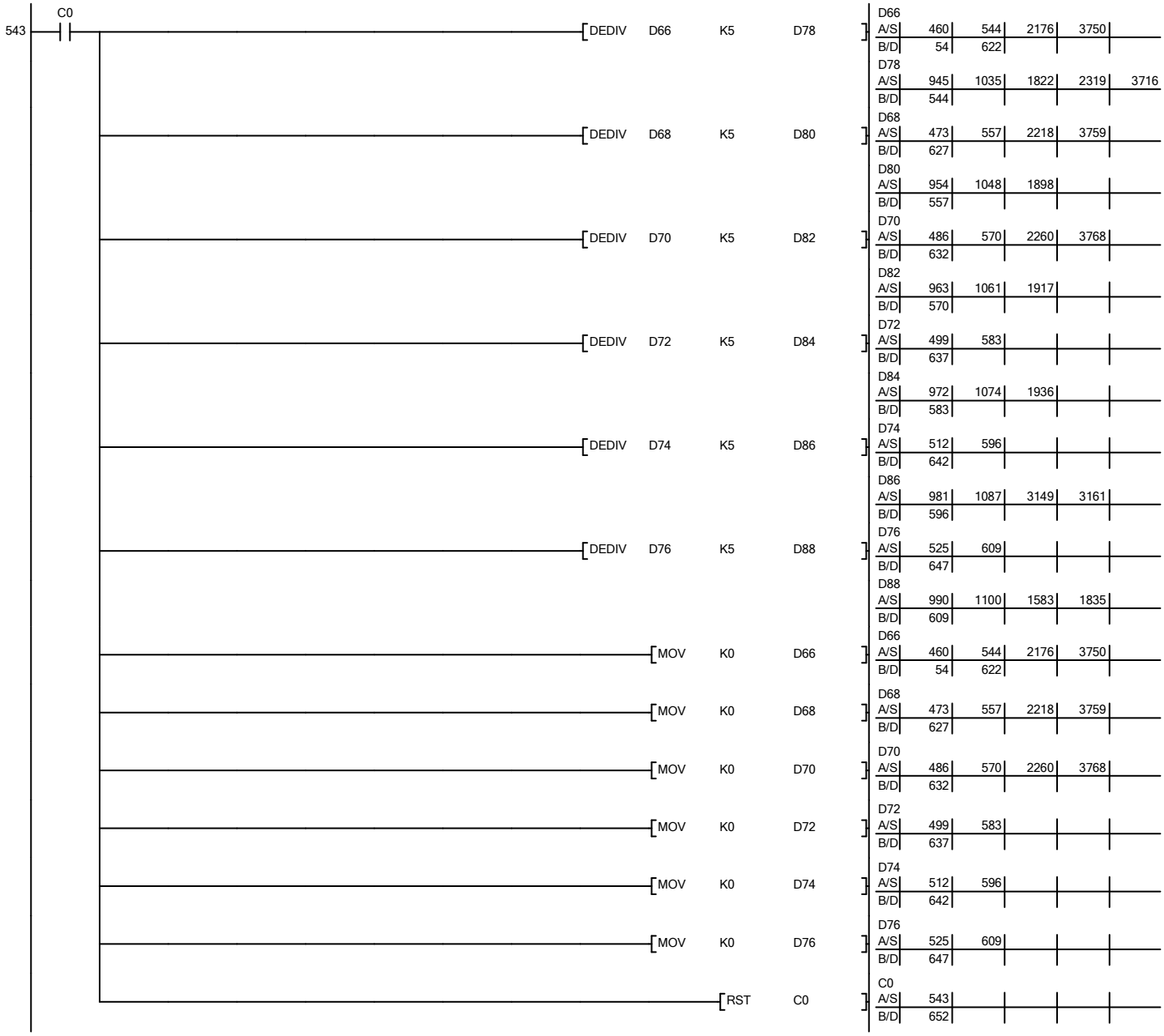

80
M8002
MOV D8030 D196
MOV D8031 D198
MOV D8231 D194
FROM K1 K2 D102 K4
FROM K2 K2 D106 K4
D8030
A/S 81
A/S
D196
A/S 115
B/D 81
D8031
A/S 86
A/S
D198
A/S 120
B/D 86
D8231
A/S 91
A/S
D194
A/S 165
B/D 91
D102
A/S 125
B/D 96
D106
A/S 145
B/D 105

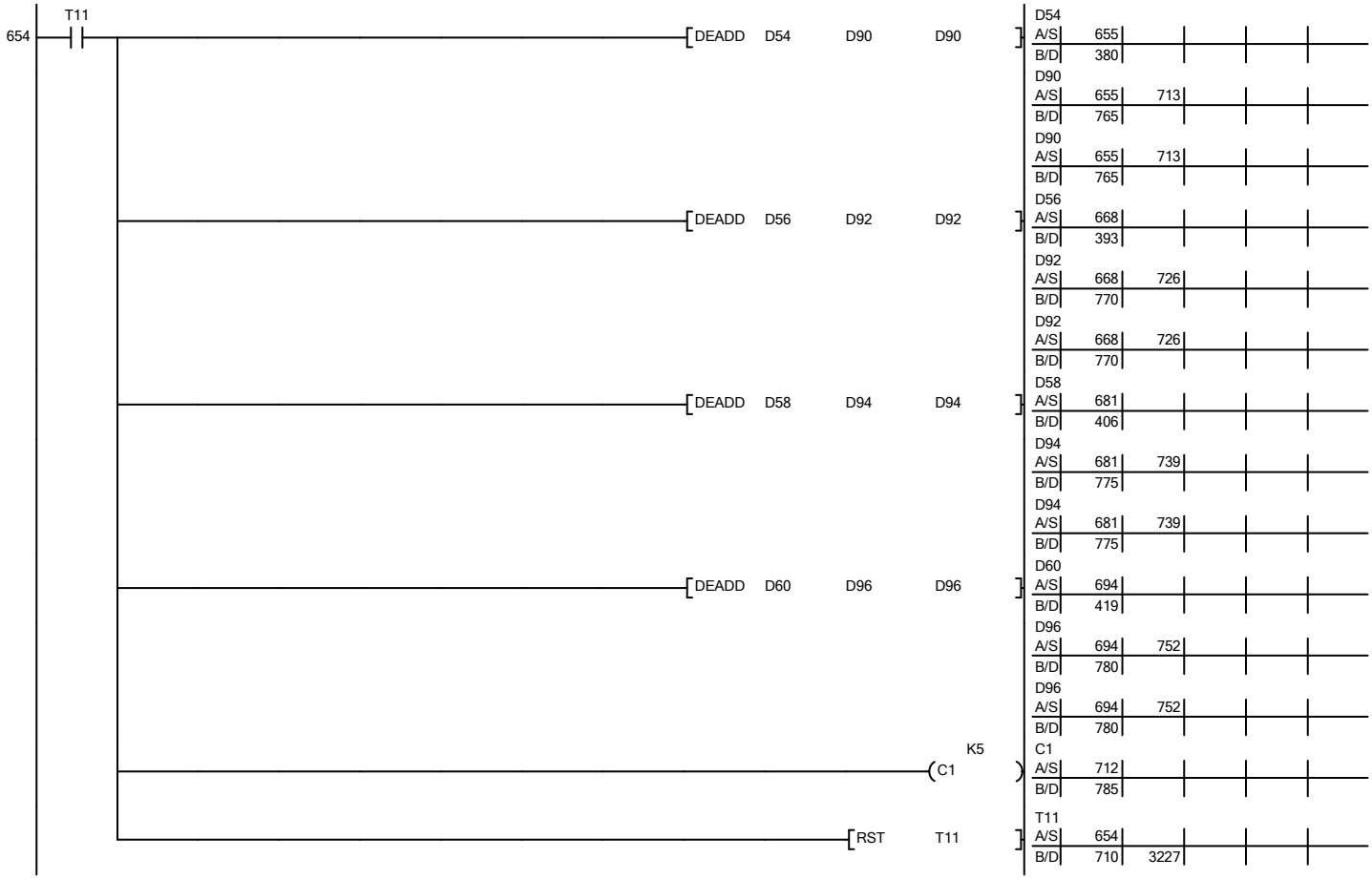

114
T0
FLT D196 D0
FLT D198 D2
FLT D102 D4
FLT D103 D6
FLT D104 D8
FLT D105 D10
FLT D106 D12
FLT D107 D14
FLT D108 D16
FLT D109 D18
FLT D194 D20
D196 A/S 115 B/D 81
D0 A/S 171 B/D 115
D198 A/S 120 B/D 86
D2 A/S 184 B/D 120
D102 A/S 125 B/D 96
D4 A/S 197 B/D 125
D103 A/S 130 A/S
D6 A/S 210 B/D 130
D104 A/S 135 A/S
D8 A/S 223 B/D 135
D105 A/S 140 A/S
D10 A/S 236 B/D 140
D106 A/S 145 B/D 105
D12 A/S 328 B/D 145
D107 A/S 150 A/S
D14 A/S 341 B/D 150
D108 A/S 155 A/S
D16 A/S 354 B/D 155
D109 A/S 160 A/S
D18 A/S 367 B/D 160
D194 A/S 165 B/D 91
D20 A/S 433 B/D 165

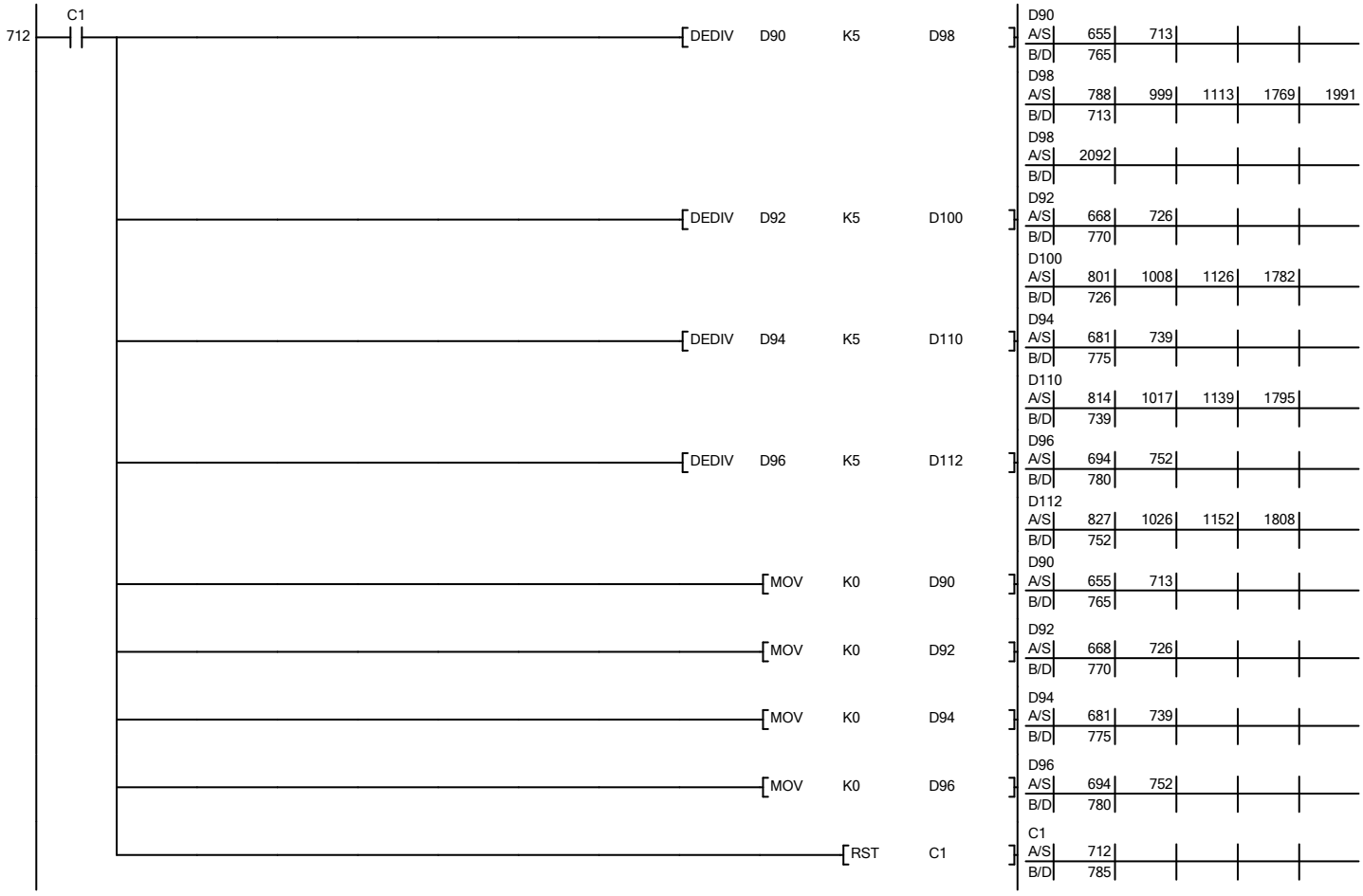

170
T0
DEMUL D0 E2.5 D22
D0 A/S 171 B/D 115
D22 A/S 249 B/D 171
DEMUL D2 E2.5 D24
D2 A/S 184 B/D 120
D24 A/S 262 B/D 184
DEMUL D4 E2.5 D26
D4 A/S 197 B/D 125
D26 A/S 275 B/D 197
DEMUL D6 E2.6 D28
D6 A/S 210 B/D 130
D28 A/S 288 B/D 210
DEMUL D8 E2.5 D30
D8 A/S 223 B/D 135
D30 A/S 301 B/D 223
DEMUL D10 E2.5 D32
D10 A/S 236 B/D 140
D32 A/S 314 B/D 236
DESUB D22 E1 D34
D22 A/S 249 B/D 171
D34 A/S 460 B/D 249
DESUB D24 E1 D36
D24 A/S 262 B/D 184
D36 A/S 473 B/D 262
DESUB D26 E1 D38
D26 A/S 275 B/D 197
D38 A/S 486 B/D 275
DESUB D28 E1 D40
D28 A/S 288 B/D 210
D40 A/S 499 B/D 288
DESUB D30 E1 D42
D30 A/S 301 B/D 223
D42 A/S 512 B/D 301
DESUB D32 E1 D44
D32 A/S 314 B/D 236
D44 A/S 525 B/D 314

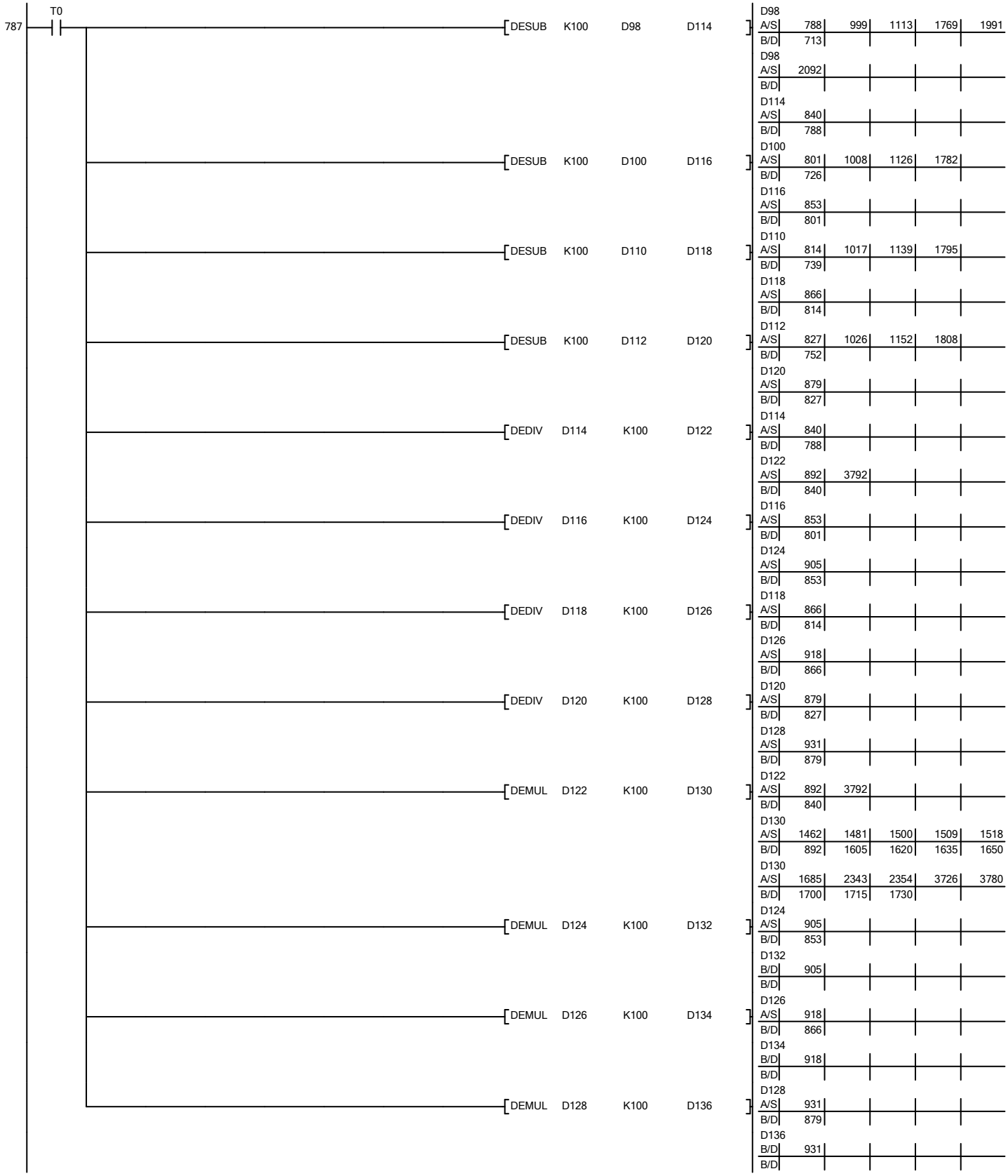

327
T0
DEMUL D12 E2.5 D46
DEMUL D14 E2.5 D48
DEMUL D16 E2.5 D50
DEMUL D18 E2.5 D52
DESUB D46 E1 D54
DESUB D48 E1 D56
DESUB D50 E1 D58
DESUB D52 E1 D60
432
T0
DEMUL D20 E2.5 D62
DESUB D62 E1 D64
D12 A/S 328 B/D 145
D46 A/S 380 B/D 328
D14 A/S 341 B/D 150
D48 A/S 393 B/D 341
D16 A/S 354 B/D 155
D50 A/S 406 B/D 354
D18 A/S 367 B/D 160
D52 A/S 419 B/D 367
D46 A/S 380 B/D 328
D54 A/S 655 B/D 380
D48 A/S 393 B/D 341
D56 A/S 668 B/D 393
D50 A/S 406 B/D 354
D58 A/S 681 B/D 406
D52 A/S 419 B/D 367
D60 A/S 694 B/D 419
D20 A/S 433 B/D 165
D62 A/S 446 B/D 433
D62 A/S 446 B/D 433
D64 A/S 3420 B/D 446 3405

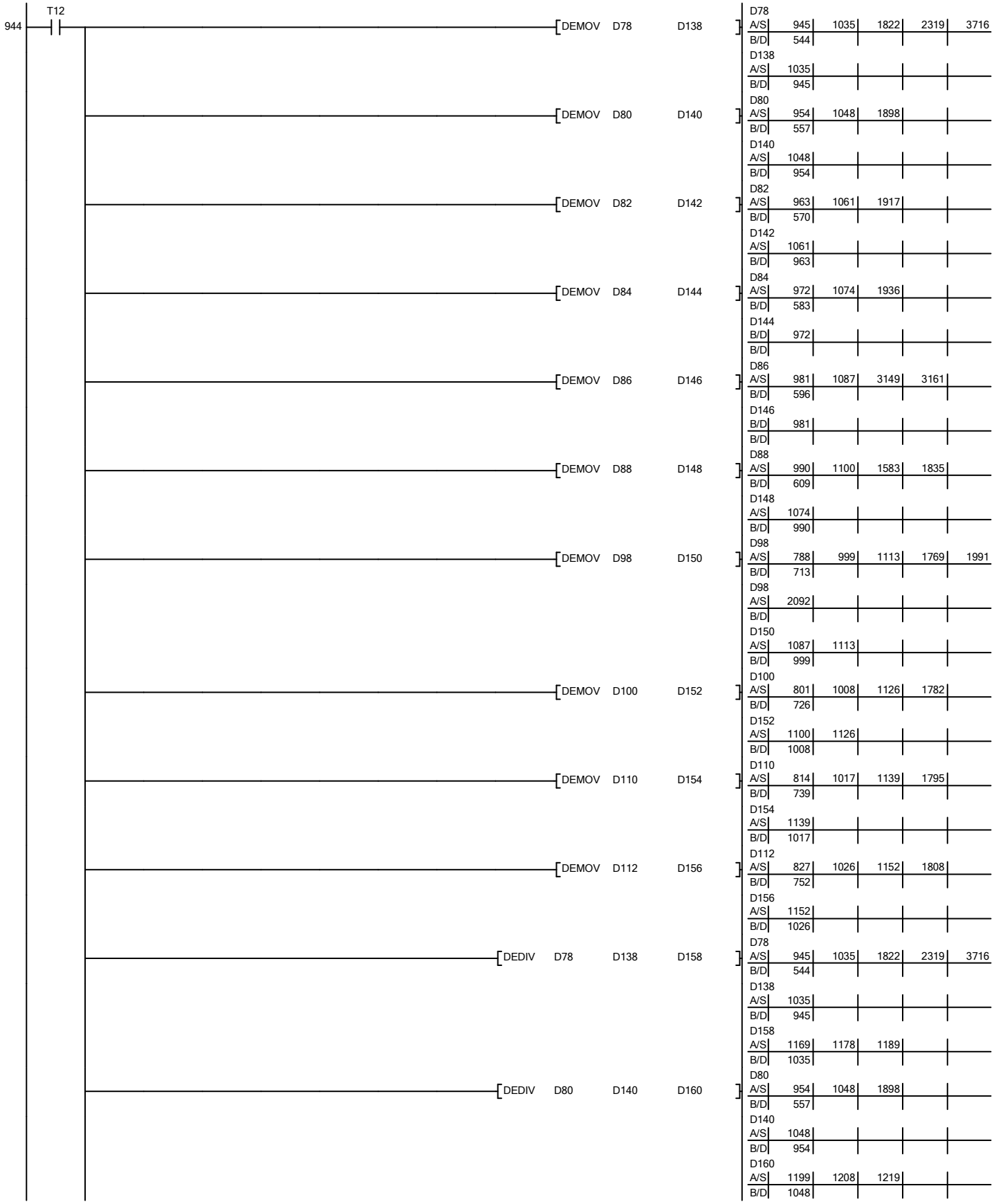

T10
459
DEADD D34 D66 D66
DEADD D36 D68 D68
DEADD D38 D70 D70
DEADD D40 D72 D72
DEADD D42 D74 D74
DEADD D44 D76 D76
K5
C0
RST T10
D34 A/S 460 B/D 249
D66 A/S 460 544 2176 3750 B/D 54 622
D66 A/S 460 544 2176 3750 B/D 54 622
D36 A/S 473 B/D 262
D68 A/S 473 557 2218 3759 B/D 627
D68 A/S 473 557 2218 3759 B/D 627
D38 A/S 486 B/D 275
D70 A/S 486 570 2260 3768 B/D 632
D70 A/S 486 570 2260 3768 B/D 632
D40 A/S 499 B/D 288
D72 A/S 499 583 B/D 637
D72 A/S 499 583 B/D 637
D42 A/S 512 B/D 301
D74 A/S 512 596 B/D 642
D74 A/S 512 596 B/D 642
D44 A/S 525 B/D 314
D76 A/S 525 609 B/D 647
D76 A/S 525 609 B/D 647
C0 A/S 543 B/D 652
T10 A/S 459 B/D 541 3225

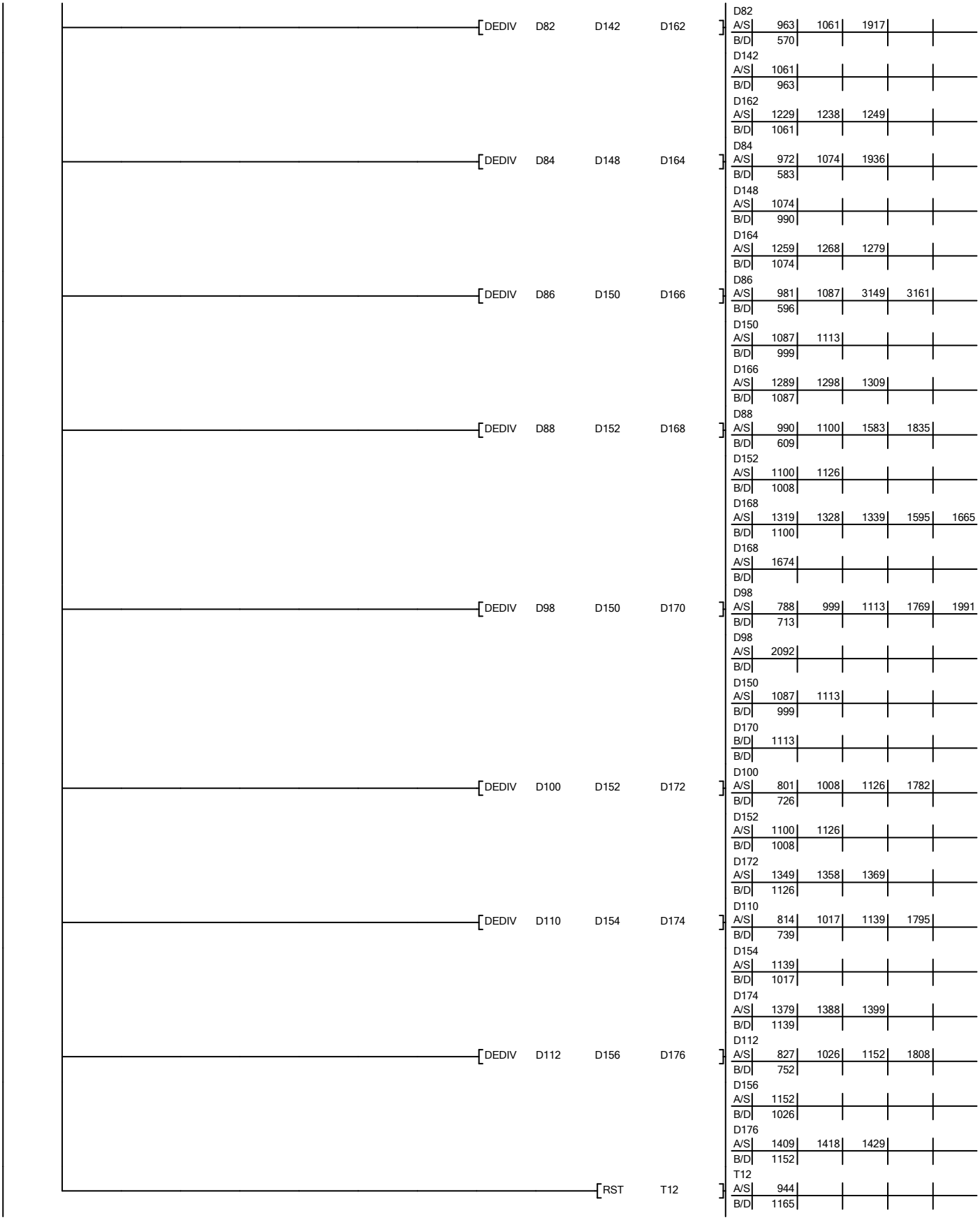

C0
543
DEDIV D66 K5 D78
DEDIV D68 K5 D80
DEDIV D70 K5 D82
DEDIV D72 K5 D84
DEDIV D74 K5 D86
DEDIV D76 K5 D88
MOV K0 D66
MOV K0 D68
MOV K0 D70
MOV K0 D72
MOV K0 D74
MOV K0 D76
RST C0
D66 A/S 460 544 2176 3750 B/D 54 622
D78 A/S 945 1035 1822 2319 3716 B/D 544
D68 A/S 473 557 2218 3759 B/D 627
D80 A/S 954 1048 1898 B/D 557
D70 A/S 486 570 2260 3768 B/D 632
D82 A/S 963 1061 1917 B/D 570
D72 A/S 499 583 B/D 637
D84 A/S 972 1074 1936 B/D 583
D74 A/S 512 596 B/D 642
D86 A/S 981 1087 3149 3161 B/D 596
D76 A/S 525 609 B/D 647
D88 A/S 990 1100 1583 1835 B/D 609
D66 A/S 460 544 2176 3750 B/D 54 622
D68 A/S 473 557 2218 3759 B/D 627
D70 A/S 486 570 2260 3768 B/D 632
D72 A/S 499 583 B/D 637
D74 A/S 512 596 B/D 642
D76 A/S 525 609 B/D 647
C0 A/S 543 B/D 652

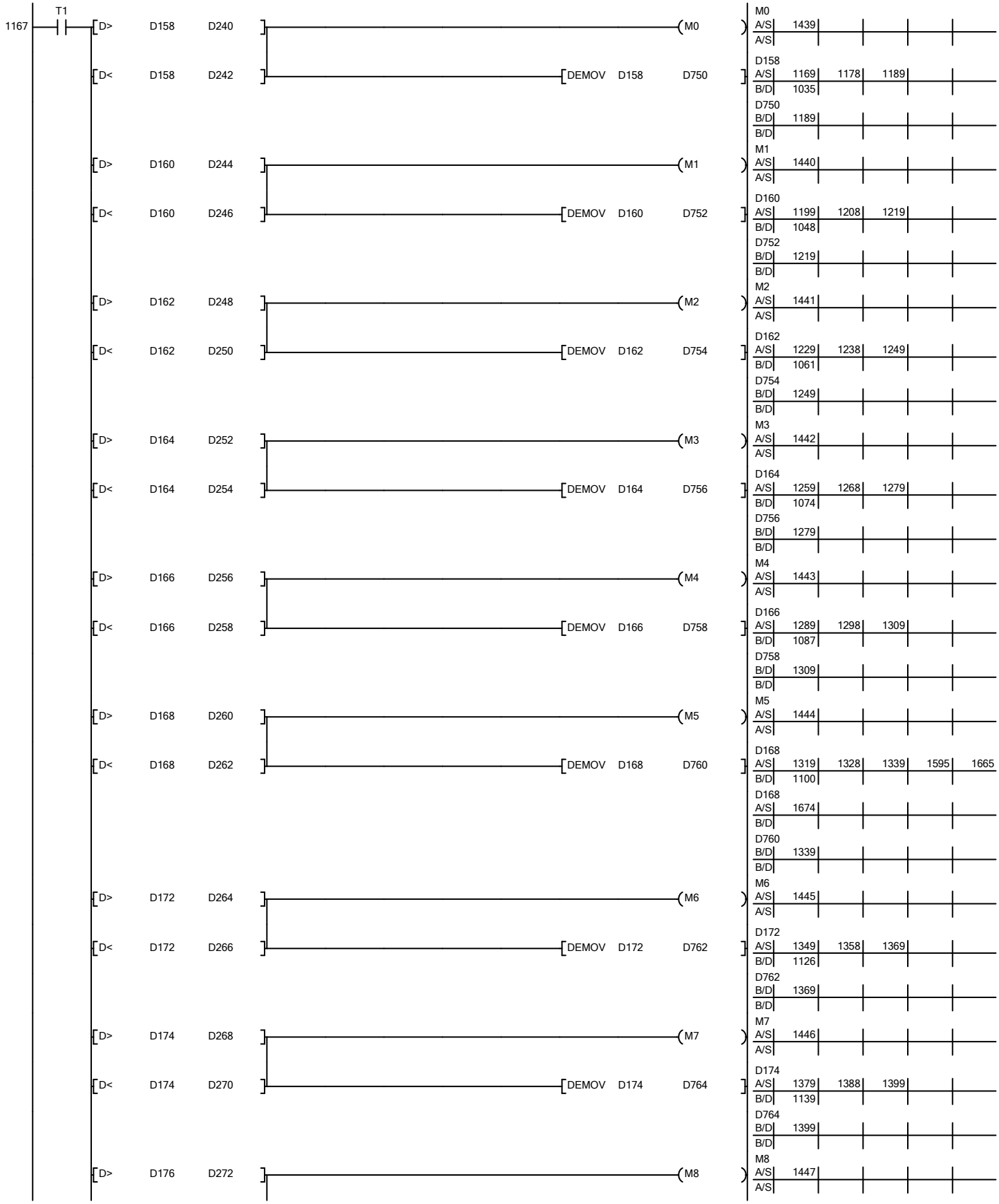

654
T11
DEADD D54 D90 D90
DEADD D56 D92 D92
DEADD D58 D94 D94
DEADD D60 D96 D96
K5
C1
RST T11
D54 A/S 655 B/D 380
D90 A/S 655 713 B/D 765
D90 A/S 655 713 B/D 765
D56 A/S 668 B/D 393
D92 A/S 668 726 B/D 770
D92 A/S 668 726 B/D 770
D58 A/S 681 B/D 406
D94 A/S 681 739 B/D 775
D94 A/S 681 739 B/D 775
D60 A/S 694 B/D 419
D96 A/S 694 752 B/D 780
D96 A/S 694 752 B/D 780
C1 A/S 712 B/D 785
T11 A/S 654 B/D 710 3227

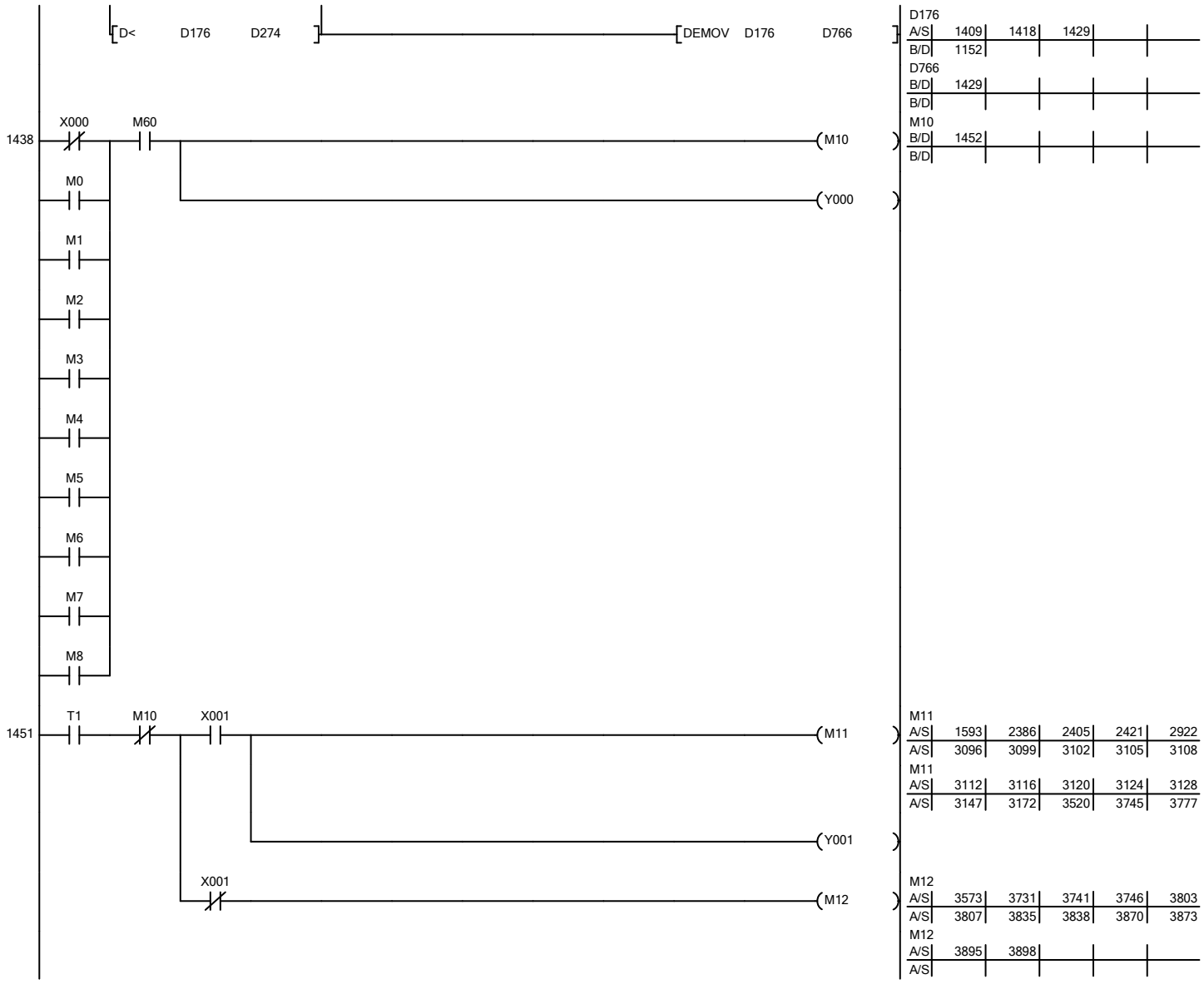

C1
712
DEDIV D90 K5 D98
DEDIV D92 K5 D100
DEDIV D94 K5 D110
DEDIV D96 K5 D112
MOV K0 D90
MOV K0 D92
MOV K0 D94
MOV K0 D96
RST C1
D90 A/S 655 713 B/D 765
D98 A/S 788 999 1113 1769 1991 B/D 713
D98 A/S 2092 B/D
D92 A/S 668 726 B/D 770
D100 A/S 801 1008 1126 1782 B/D 726
D94 A/S 681 739 B/D 775
D110 A/S 814 1017 1139 1795 B/D 739
D96 A/S 694 752 B/D 780
D112 A/S 827 1026 1152 1808 B/D 752
D90 A/S 655 713 B/D 765
D92 A/S 668 726 B/D 770
D94 A/S 681 739 B/D 775
D96 A/S 694 752 B/D 780
C1 A/S 712 B/D 785

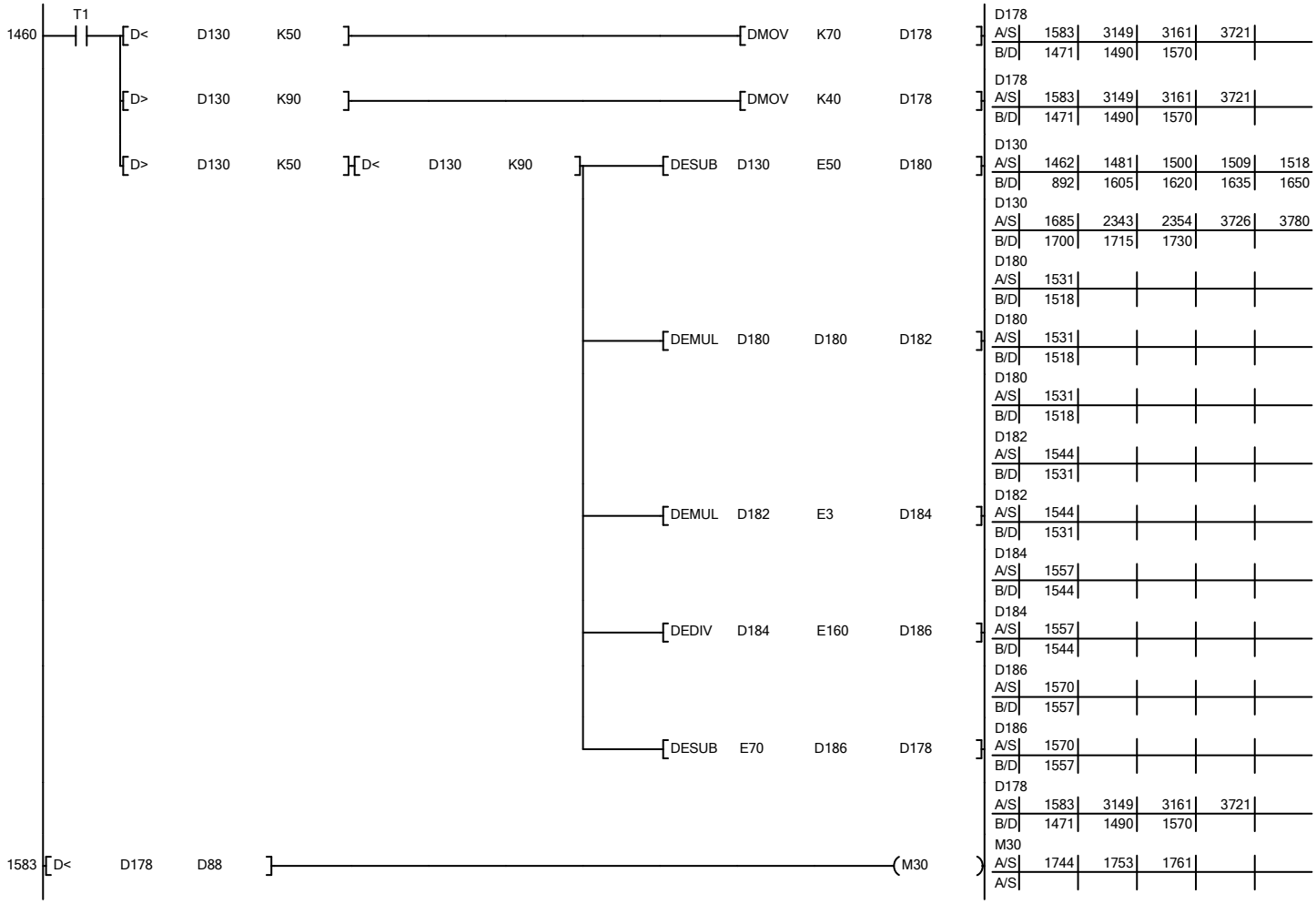

787
T0
DESUB K100 D98 D114
DESUB K100 D100 D116
DESUB K100 D110 D118
DESUB K100 D112 D120
DEDIV D114 K100 D122
DEDIV D116 K100 D124
DEDIV D118 K100 D126
DEDIV D120 K100 D128
DEMUL D122 K100 D130
DEMUL D124 K100 D132
DEMUL D126 K100 D134
DEMUL D128 K100 D136
D98 A/S 788 999 1113 1769 1991
B/D 713
D98 A/S 2092
B/D
D114 A/S 840
B/D 788
D100 A/S 801 1008 1126 1782
B/D 726
D116 A/S 853
B/D 801
D110 A/S 814 1017 1139 1795
B/D 739
D118 A/S 866
B/D 814
D112 A/S 827 1026 1152 1808
B/D 752
D120 A/S 879
B/D 827
D114 A/S 840
B/D 788
D122 A/S 892 3792
B/D 840
D116 A/S 853
B/D 801
D124 A/S 905
B/D 853
D118 A/S 866
B/D 814
D126 A/S 918
B/D 866
D120 A/S 879
B/D 827
D128 A/S 931
B/D 879
D122 A/S 892 3792
B/D 840
D130 A/S 1462 1481 1500 1509 1518
B/D 892 1605 1620 1635 1650
D130 A/S 1685 2343 2354 3726 3780
B/D 1700 1715 1730
D124 A/S 905
B/D 853
D132 B/D 905
B/D
D126 A/S 918
B/D 866
D134 B/D 918
B/D
D128 A/S 931
B/D 879
D136 B/D 931
B/D

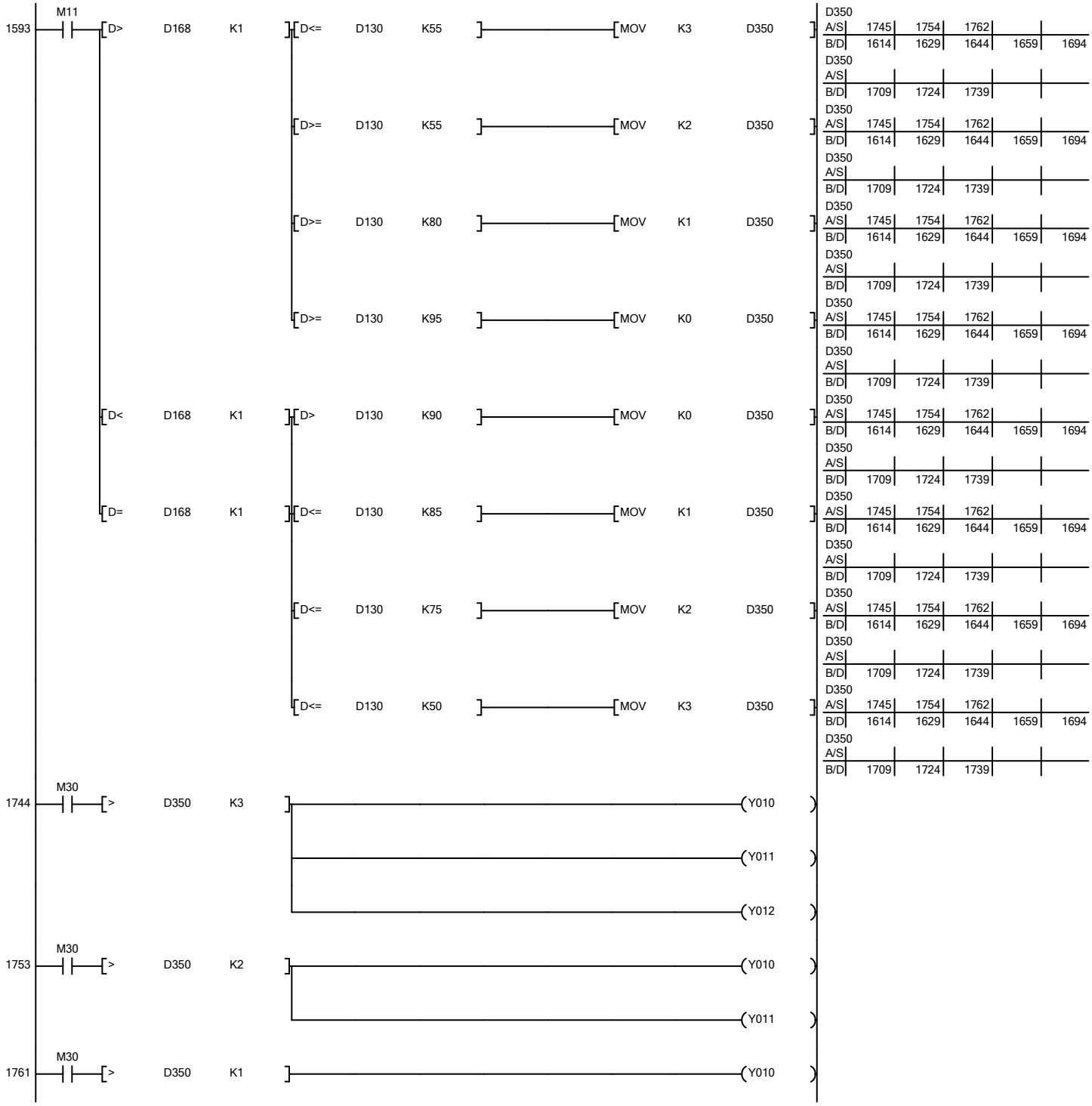

944
T12
DEMOV D78 D138
DEMOV D80 D140
DEMOV D82 D142
DEMOV D84 D144
DEMOV D86 D146
DEMOV D88 D148
DEMOV D98 D150
DEMOV D100 D152
DEMOV D110 D154
DEMOV D112 D156
DEDIV D78 D138 D158
DEDIV D80 D140 D160
D78 A/S 945 1035 1822 2319 3716 B/D 544
D138 A/S 1035 B/D 945
D80 A/S 954 1048 1898 B/D 557
D140 A/S 1048 B/D 954
D82 A/S 963 1061 1917 B/D 570
D142 A/S 1061 B/D 963
D84 A/S 972 1074 1936 B/D 583
D144 B/D 972 B/D
D86 A/S 981 1087 3149 3161 B/D 596
D146 B/D 981 B/D
D88 A/S 990 1100 1583 1835 B/D 609
D148 A/S 1074 B/D 990
D98 A/S 788 999 1113 1769 1991 B/D 713
D98 A/S 2092 B/D
D150 A/S 1087 1113 B/D 999
D100 A/S 801 1008 1126 1782 B/D 726
D152 A/S 1100 1126 B/D 1008
D110 A/S 814 1017 1139 1795 B/D 739
D154 A/S 1139 B/D 1017
D112 A/S 827 1026 1152 1808 B/D 752
D156 A/S 1152 B/D 1026
D78 A/S 945 1035 1822 2319 3716 B/D 544
D138 A/S 1035 B/D 945
D158 A/S 1169 1178 1189 B/D 1035
D80 A/S 954 1048 1898 B/D 557
D140 A/S 1048 B/D 954
D160 A/S 1199 1208 1219 B/D 1048

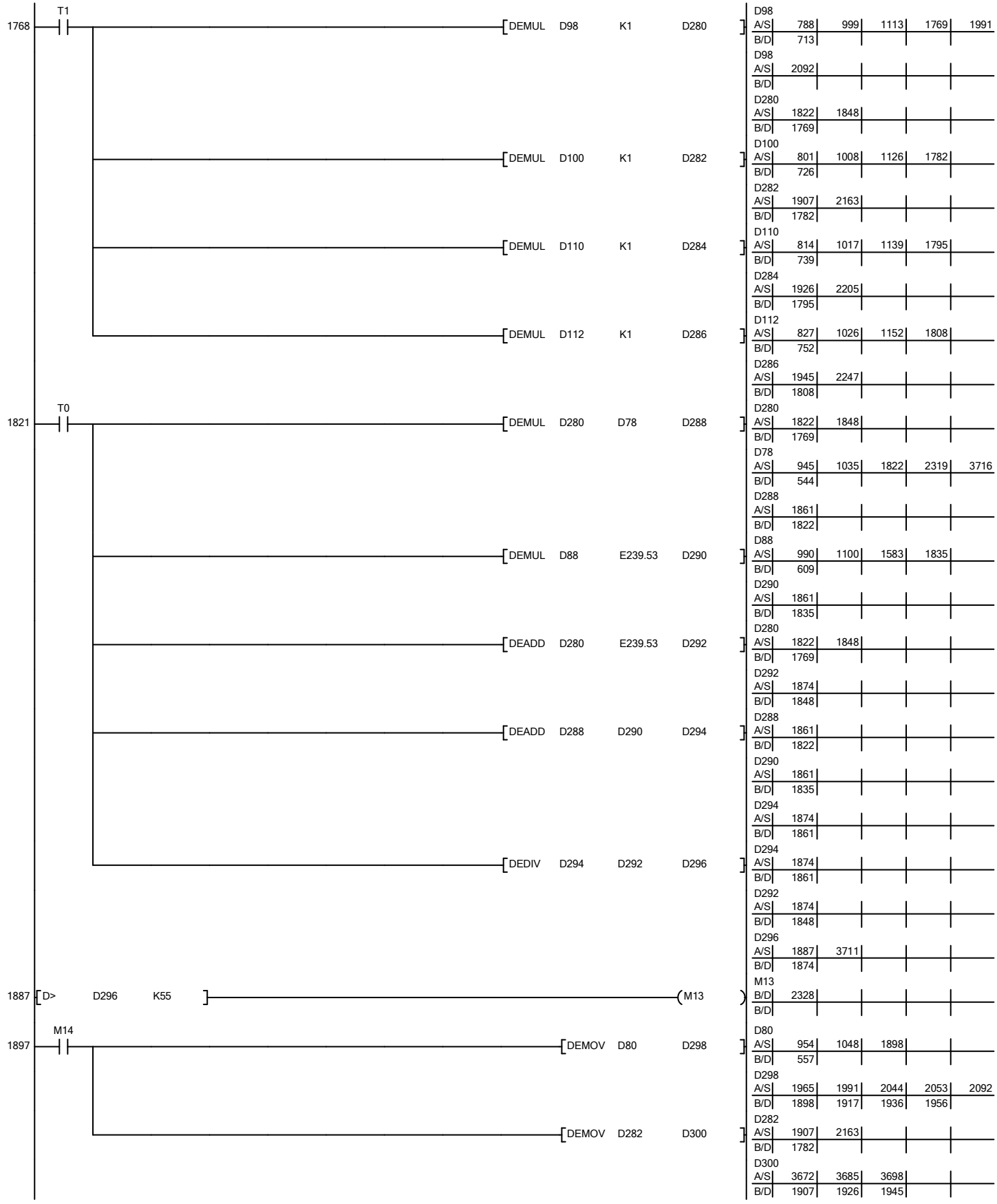

DEDIV D82 D142 D162
D82 A/S 963 1061 1917 B/D 570
D142 A/S 1061 B/D 963
D162 A/S 1229 1238 1249 B/D 1061
DEDIV D84 D148 D164
D84 A/S 972 1074 1936 B/D 583
D148 A/S 1074 B/D 990
D164 A/S 1259 1268 1279 B/D 1074
DEDIV D86 D150 D166
D86 A/S 981 1087 3149 3161 B/D 596
D150 A/S 1087 1113 B/D 999
D166 A/S 1289 1298 1309 B/D 1087
DEDIV D88 D152 D168
D88 A/S 990 1100 1583 1835 B/D 609
D152 A/S 1100 1126 B/D 1008
D168 A/S 1319 1328 1339 1595 1665 B/D 1100
D168 A/S 1674 B/D
DEDIV D98 D150 D170
D98 A/S 788 999 1113 1769 1991 B/D 713
D98 A/S 2092 B/D
D150 A/S 1087 1113 B/D 999
D170 B/D 1113 B/D
DEDIV D100 D152 D172
D100 A/S 801 1008 1126 1782 B/D 726
D152 A/S 1100 1126 B/D 1008
D172 A/S 1349 1358 1369 B/D 1126
DEDIV D110 D154 D174
D110 A/S 814 1017 1139 1795 B/D 739
D154 A/S 1139 B/D 1017
D174 A/S 1379 1388 1399 B/D 1139
DEDIV D112 D156 D176
D112 A/S 827 1026 1152 1808 B/D 752
D156 A/S 1152 B/D 1026
D176 A/S 1409 1418 1429 B/D 1152
RST T12
T12 A/S 944 B/D 1165


1167
T1
D> D158 D240
M0
D< D158 D242
DEMOV D158 D750
D> D160 D244
M1
D< D160 D246
DEMOV D160 D752
D> D162 D248
M2
D< D162 D250
DEMOV D162 D754
D> D164 D252
M3
D< D164 D254
DEMOV D164 D756
D> D166 D256
M4
D< D166 D258
DEMOV D166 D758
D> D168 D260
M5
D< D168 D262
DEMOV D168 D760
D> D172 D264
M6
D< D172 D266
DEMOV D172 D762
D> D174 D268
M7
D< D174 D270
DEMOV D174 D764
D> D176 D272
M8
M0 A/S 1439 A/S
D158 A/S 1169 1178 1189 B/D 1035
D750 B/D 1189 B/D
M1 A/S 1440 A/S
D160 A/S 1199 1208 1219 B/D 1048
D752 B/D 1219 B/D
M2 A/S 1441 A/S
D162 A/S 1229 1238 1249 B/D 1061
D754 B/D 1249 B/D
M3 A/S 1442 A/S
D164 A/S 1259 1268 1279 B/D 1074
D756 B/D 1279 B/D
M4 A/S 1443 A/S
D166 A/S 1289 1298 1309 B/D 1087
D758 B/D 1309 B/D
M5 A/S 1444 A/S
D168 A/S 1319 1328 1339 1595 1665 B/D 1100
D168 A/S 1674 B/D
D760 B/D 1339 B/D
M6 A/S 1445 A/S
D172 A/S 1349 1358 1369 B/D 1126
D762 B/D 1369 B/D
M7 A/S 1446 A/S
D174 A/S 1379 1388 1399 B/D 1139
D764 B/D 1399 B/D
M8 A/S 1447 A/S

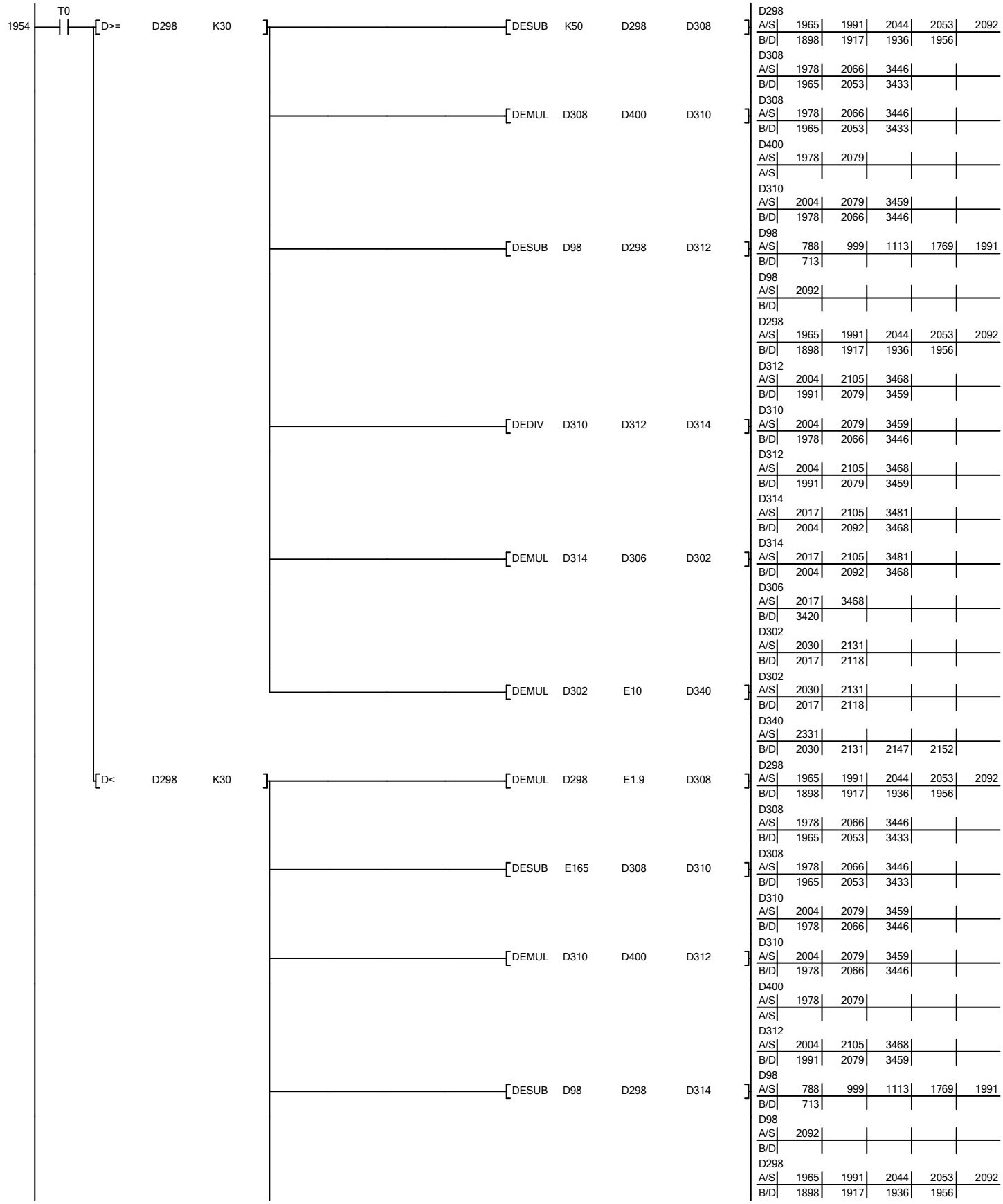

D<
D176
D274
DEMOV
D176
D766
D176
A/S 1409 1418 1429
B/D 1152
D766
B/D 1429
B/D
M10
B/D 1452
B/D
1438
X000
M60
M10
Y000
M0
M1
M2
M3
M4
M5
M6
M7
M8
1451
T1
M10
X001
M11
Y001
X001
M12
M11
A/S 1593 2386 2405 2421 2922
A/S 3096 3099 3102 3105 3108
M11
A/S 3112 3116 3120 3124 3128
A/S 3147 3172 3520 3745 3777
M12
A/S 3573 3731 3741 3746 3803
A/S 3807 3835 3838 3870 3873
M12
A/S 3895 3898
A/S

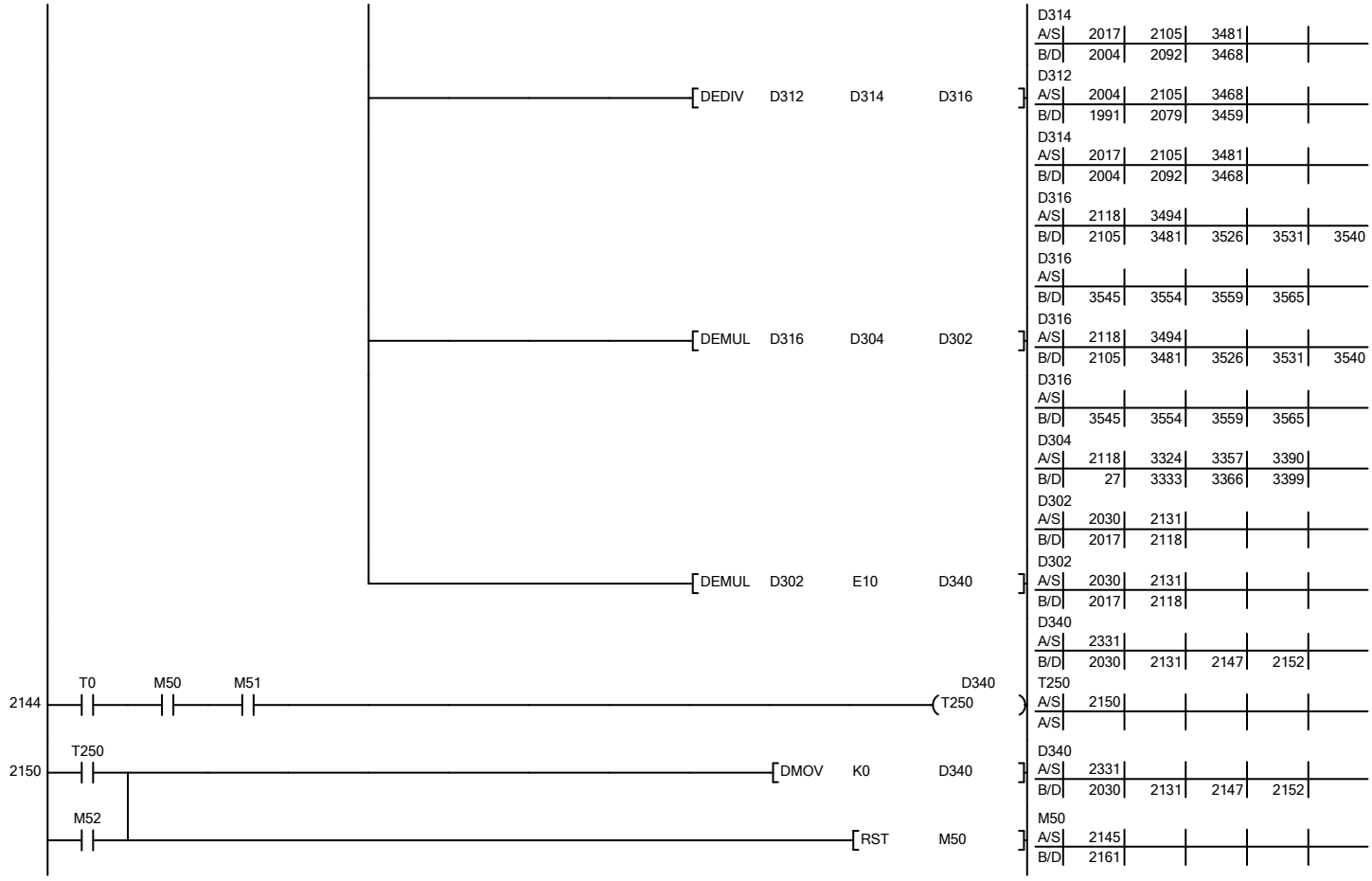

1460
T1
D< D130 K50
DMOV K70 D178
D178 A/S 1583 3149 3161 3721 B/D 1471 1490 1570
D> D130 K90
DMOV K40 D178
D178 A/S 1583 3149 3161 3721 B/D 1471 1490 1570
D> D130 K50
D< D130 K90
DESUB D130 E50 D180
D130 A/S 1462 1481 1500 1509 1518 B/D 892 1605 1620 1635 1650
D130 A/S 1685 2343 2354 3726 3780 B/D 1700 1715 1730
D180 A/S 1531 B/D 1518
DEMUL D180 D180 D182
D180 A/S 1531 B/D 1518
D180 A/S 1531 B/D 1518
D182 A/S 1544 B/D 1531
DEMUL D182 E3 D184
D182 A/S 1544 B/D 1531
D184 A/S 1557 B/D 1544
DEDIV D184 E160 D186
D184 A/S 1557 B/D 1544
D186 A/S 1570 B/D 1557
DESUB E70 D186 D178
D186 A/S 1570 B/D 1557
D178 A/S 1583 3149 3161 3721 B/D 1471 1490 1570
1583
D< D178 D88
M30
M30 A/S 1744 1753 1761 A/S

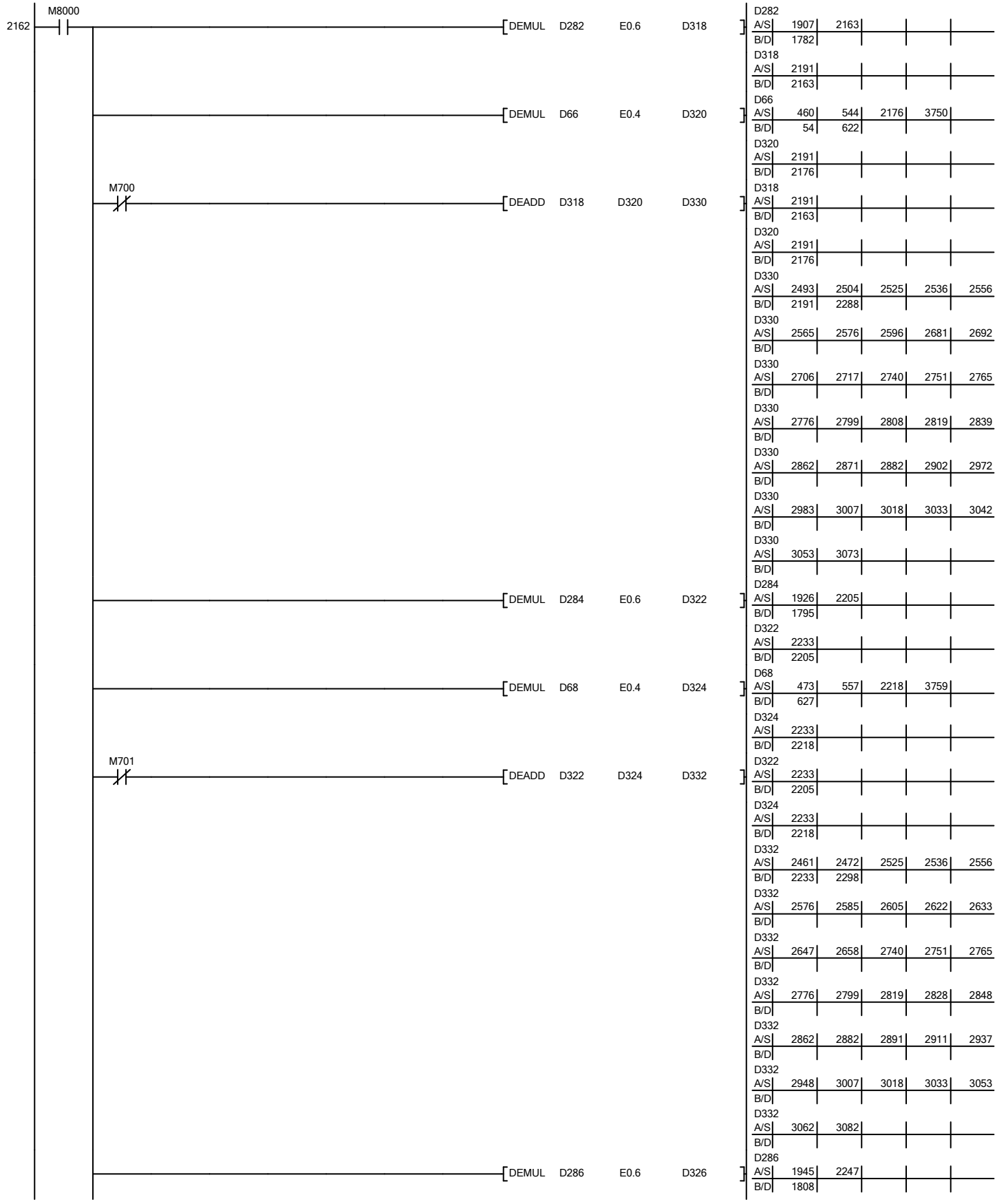

1593 M11 D> D168 K1
D<= D130 K55 MOV K3 D350
D350 A/S 1745 1754 1762 B/D 1614 1629 1644 1659 1694
D350 A/S B/D 1709 1724 1739
D>= D130 K55 MOV K2 D350
D350 A/S 1745 1754 1762 B/D 1614 1629 1644 1659 1694
D350 A/S B/D 1709 1724 1739
D>= D130 K80 MOV K1 D350
D350 A/S 1745 1754 1762 B/D 1614 1629 1644 1659 1694
D350 A/S B/D 1709 1724 1739
D>= D130 K95 MOV K0 D350
D350 A/S 1745 1754 1762 B/D 1614 1629 1644 1659 1694
D350 A/S B/D 1709 1724 1739
D< D168 K1
D> D130 K90 MOV K0 D350
D350 A/S 1745 1754 1762 B/D 1614 1629 1644 1659 1694
D350 A/S B/D 1709 1724 1739
D= D168 K1
D<= D130 K85 MOV K1 D350
D350 A/S 1745 1754 1762 B/D 1614 1629 1644 1659 1694
D350 A/S B/D 1709 1724 1739
D<= D130 K75 MOV K2 D350
D350 A/S 1745 1754 1762 B/D 1614 1629 1644 1659 1694
D350 A/S B/D 1709 1724 1739
D<= D130 K50 MOV K3 D350
D350 A/S 1745 1754 1762 B/D 1614 1629 1644 1659 1694
D350 A/S B/D 1709 1724 1739
1744 M30 > D350 K3 Y010
Y011
Y012
1753 M30 > D350 K2 Y010
Y011
1761 M30 > D350 K1 Y010

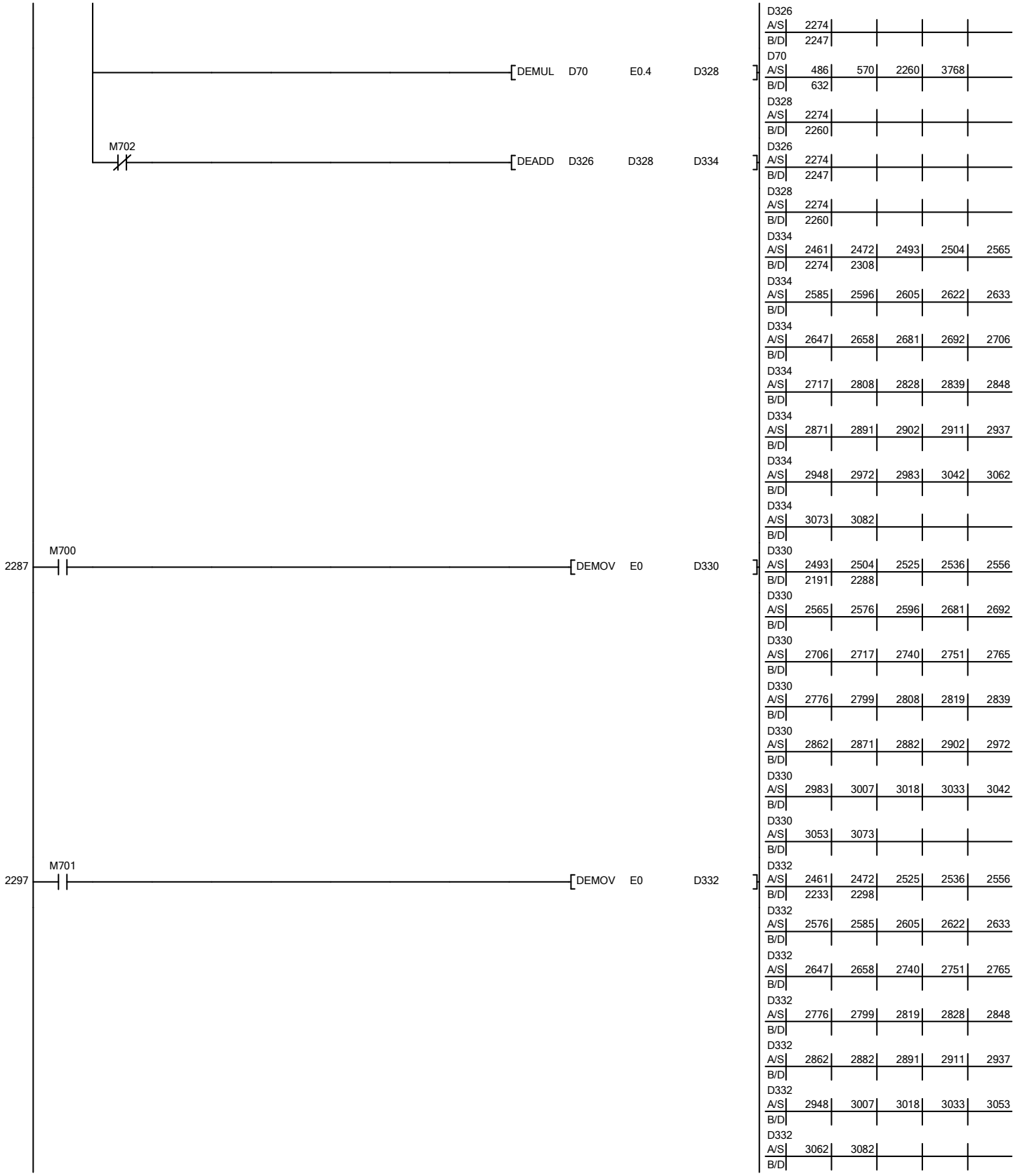

1768
T1
DEMUL D98 K1 D280
DEMUL D100 K1 D282
DEMUL D110 K1 D284
DEMUL D112 K1 D286
1821
T0
DEMUL D280 D78 D288
DEMUL D88 E239.53 D290
DEADD D280 E239.53 D292
DEADD D288 D290 D294
DEDIV D294 D292 D296
1887
D> D296 K55
M13
1897
M14
DEMOV D80 D298
DEMOV D282 D300
D98 A/S 788 999 1113 1769 1991 B/D 713
D98 A/S 2092 B/D
D280 A/S 1822 1848 B/D 1769
D100 A/S 801 1008 1126 1782 B/D 726
D282 A/S 1907 2163 B/D 1782
D110 A/S 814 1017 1139 1795 B/D 739
D284 A/S 1926 2205 B/D 1795
D112 A/S 827 1026 1152 1808 B/D 752
D286 A/S 1945 2247 B/D 1808
D280 A/S 1822 1848 B/D 1769
D78 A/S 945 1035 1822 2319 3716 B/D 544
D288 A/S 1861 B/D 1822
D88 A/S 990 1100 1583 1835 B/D 609
D290 A/S 1861 B/D 1835
D280 A/S 1822 1848 B/D 1769
D292 A/S 1874 B/D 1848
D288 A/S 1861 B/D 1822
D290 A/S 1861 B/D 1835
D294 A/S 1874 B/D 1861
D294 A/S 1874 B/D 1861
D292 A/S 1874 B/D 1848
D296 A/S 1887 3711 B/D 1874
M13 B/D 2328 B/D
D80 A/S 954 1048 1898 B/D 557
D298 A/S 1965 1991 2044 2053 2092 B/D 1898 1917 1936 1956
D282 A/S 1907 2163 B/D 1782
D300 A/S 3672 3685 3698 B/D 1907 1926 1945

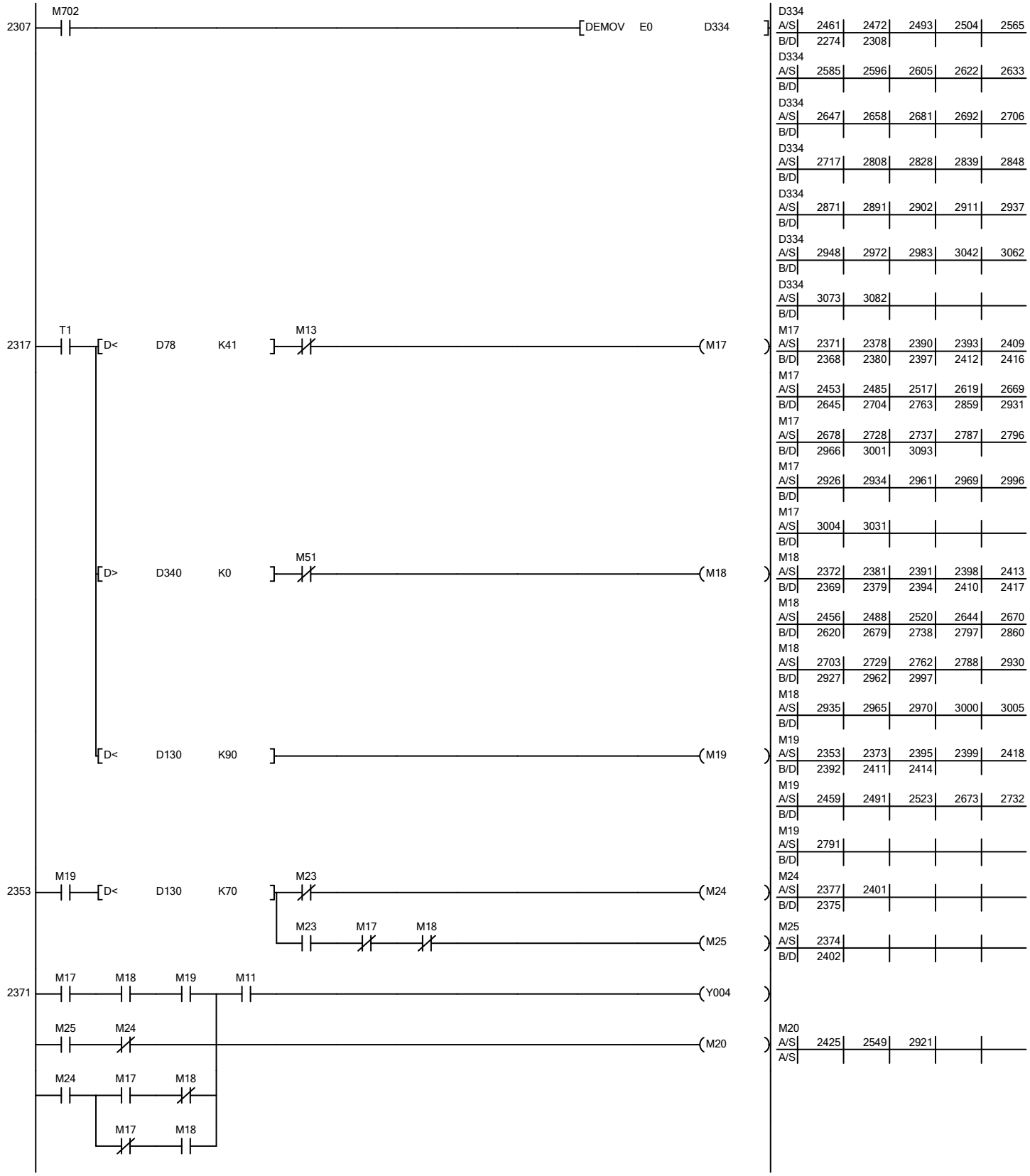

M15
1916
DEMOV D82 D298
DEMOV D284 D300
M16
1935
DEMOV D84 D298
DEMOV D286 D300
D82
A/S 963 1061 1917
B/D 570
D298
A/S 1965 1991 2044 2053 2092
B/D 1898 1917 1936 1956
D284
A/S 1926 2205
B/D 1795
D300
A/S 3672 3685 3698
B/D 1907 1926 1945
D84
A/S 972 1074 1936
B/D 583
D298
A/S 1965 1991 2044 2053 2092
B/D 1898 1917 1936 1956
D286
A/S 1945 2247
B/D 1808
D300
A/S 3672 3685 3698
B/D 1907 1926 1945


1954
T0
D>= D298 K30
DESUB K50 D298 D308
DEMUL D308 D400 D310
DESUB D98 D298 D312
DEDIV D310 D312 D314
DEMUL D314 D306 D302
DEMUL D302 E10 D340
D< D298 K30
DEMUL D298 E1.9 D308
DESUB E165 D308 D310
DEMUL D310 D400 D312
DESUB D98 D298 D314
D298 A/S 1965 1991 2044 2053 2092 B/D 1898 1917 1936 1956
D308 A/S 1978 2066 3446 B/D 1965 2053 3433
D308 A/S 1978 2066 3446 B/D 1965 2053 3433
D400 A/S 1978 2079 A/S
D310 A/S 2004 2079 3459 B/D 1978 2066 3446
D98 A/S 788 999 1113 1769 1991 B/D 713
D98 A/S 2092 B/D
D298 A/S 1965 1991 2044 2053 2092 B/D 1898 1917 1936 1956
D312 A/S 2004 2105 3468 B/D 1991 2079 3459
D310 A/S 2004 2079 3459 B/D 1978 2066 3446
D312 A/S 2004 2105 3468 B/D 1991 2079 3459
D314 A/S 2017 2105 3481 B/D 2004 2092 3468
D314 A/S 2017 2105 3481 B/D 2004 2092 3468
D306 A/S 2017 3468 B/D 3420
D302 A/S 2030 2131 B/D 2017 2118
D302 A/S 2030 2131 B/D 2017 2118
D340 A/S 2331 B/D 2030 2131 2147 2152
D298 A/S 1965 1991 2044 2053 2092 B/D 1898 1917 1936 1956
D308 A/S 1978 2066 3446 B/D 1965 2053 3433
D308 A/S 1978 2066 3446 B/D 1965 2053 3433
D310 A/S 2004 2079 3459 B/D 1978 2066 3446
D310 A/S 2004 2079 3459 B/D 1978 2066 3446
D400 A/S 1978 2079 A/S
D312 A/S 2004 2105 3468 B/D 1991 2079 3459
D98 A/S 788 999 1113 1769 1991 B/D 713
D98 A/S 2092 B/D
D298 A/S 1965 1991 2044 2053 2092 B/D 1898 1917 1936 1956

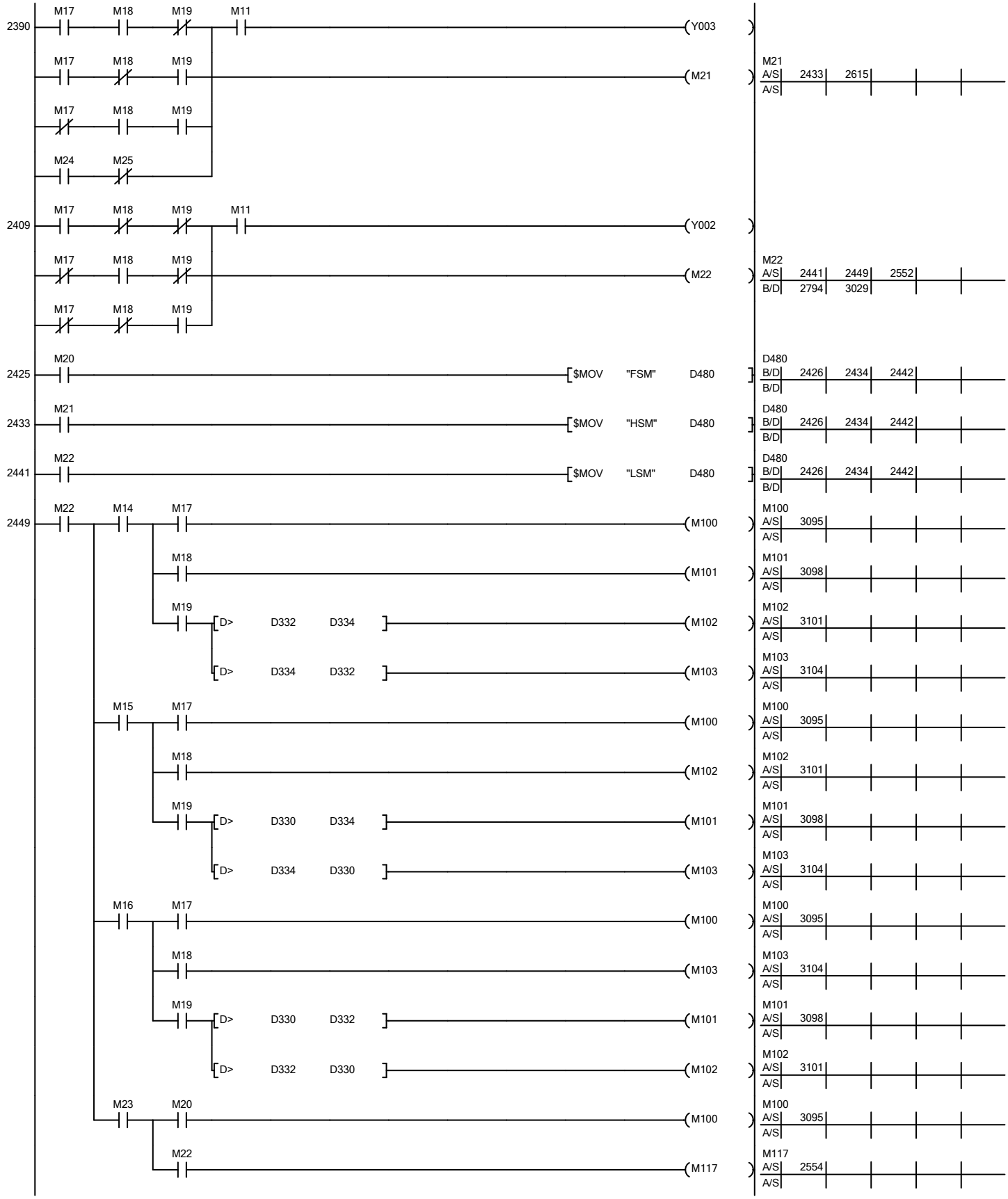

DEDIV D312 D314 D316
DEMUL D316 D304 D302
DEMUL D302 E10 D340
2144 T0 M50 M51 D340 T250
2150 T250 DMOV K0 D340
M52 RST M50
D314 A/S 2017 2105 3481 B/D 2004 2092 3468
D312 A/S 2004 2105 3468 B/D 1991 2079 3459
D314 A/S 2017 2105 3481 B/D 2004 2092 3468
D316 A/S 2118 3494 B/D 2105 3481 3526 3531 3540
D316 A/S B/D 3545 3554 3559 3565
D316 A/S 2118 3494 B/D 2105 3481 3526 3531 3540
D316 A/S B/D 3545 3554 3559 3565
D304 A/S 2118 3324 3357 3390 B/D 27 3333 3366 3399
D302 A/S 2030 2131 B/D 2017 2118
D302 A/S 2030 2131 B/D 2017 2118
D340 A/S 2331 B/D 2030 2131 2147 2152
T250 A/S 2150 A/S
D340 A/S 2331 B/D 2030 2131 2147 2152
M50 A/S 2145 B/D 2161

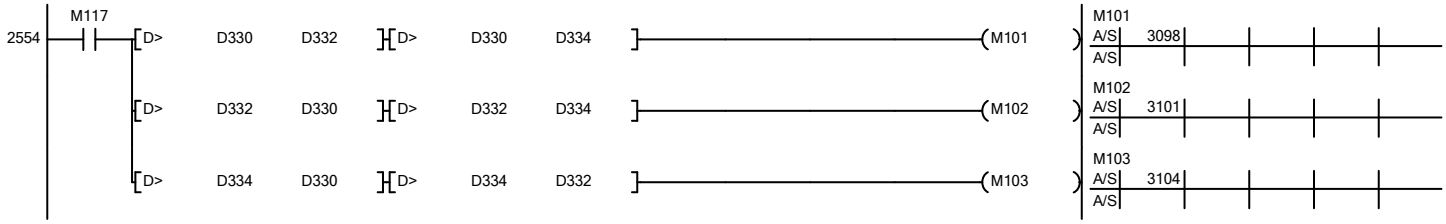

2162
M8000
DEMUL D282 E0.6 D318
D282 A/S 1907 2163 B/D 1782
D318 A/S 2191 B/D 2163
DEMUL D66 E0.4 D320
D66 A/S 460 544 2176 3750 B/D 54 622
D320 A/S 2191 B/D 2176
M700
DEADD D318 D320 D330
D318 A/S 2191 B/D 2163
D320 A/S 2191 B/D 2176
D330 A/S 2493 2504 2525 2536 2556 B/D 2191 2288
D330 A/S 2565 2576 2596 2681 2692 B/D
D330 A/S 2706 2717 2740 2751 2765 B/D
D330 A/S 2776 2799 2808 2819 2839 B/D
D330 A/S 2862 2871 2882 2902 2972 B/D
D330 A/S 2983 3007 3018 3033 3042 B/D
D330 A/S 3053 3073 B/D
DEMUL D284 E0.6 D322
D284 A/S 1926 2205 B/D 1795
D322 A/S 2233 B/D 2205
DEMUL D68 E0.4 D324
D68 A/S 473 557 2218 3759 B/D 627
D324 A/S 2233 B/D 2218
M701
DEADD D322 D324 D332
D322 A/S 2233 B/D 2205
D324 A/S 2233 B/D 2218
D332 A/S 2461 2472 2525 2536 2556 B/D 2233 2298
D332 A/S 2576 2585 2605 2622 2633 B/D
D332 A/S 2647 2658 2740 2751 2765 B/D
D332 A/S 2776 2799 2819 2828 2848 B/D
D332 A/S 2862 2882 2891 2911 2937 B/D
D332 A/S 2948 3007 3018 3033 3053 B/D
D332 A/S 3062 3082 B/D
DEMUL D286 E0.6 D326
D286 A/S 1945 2247 B/D 1808

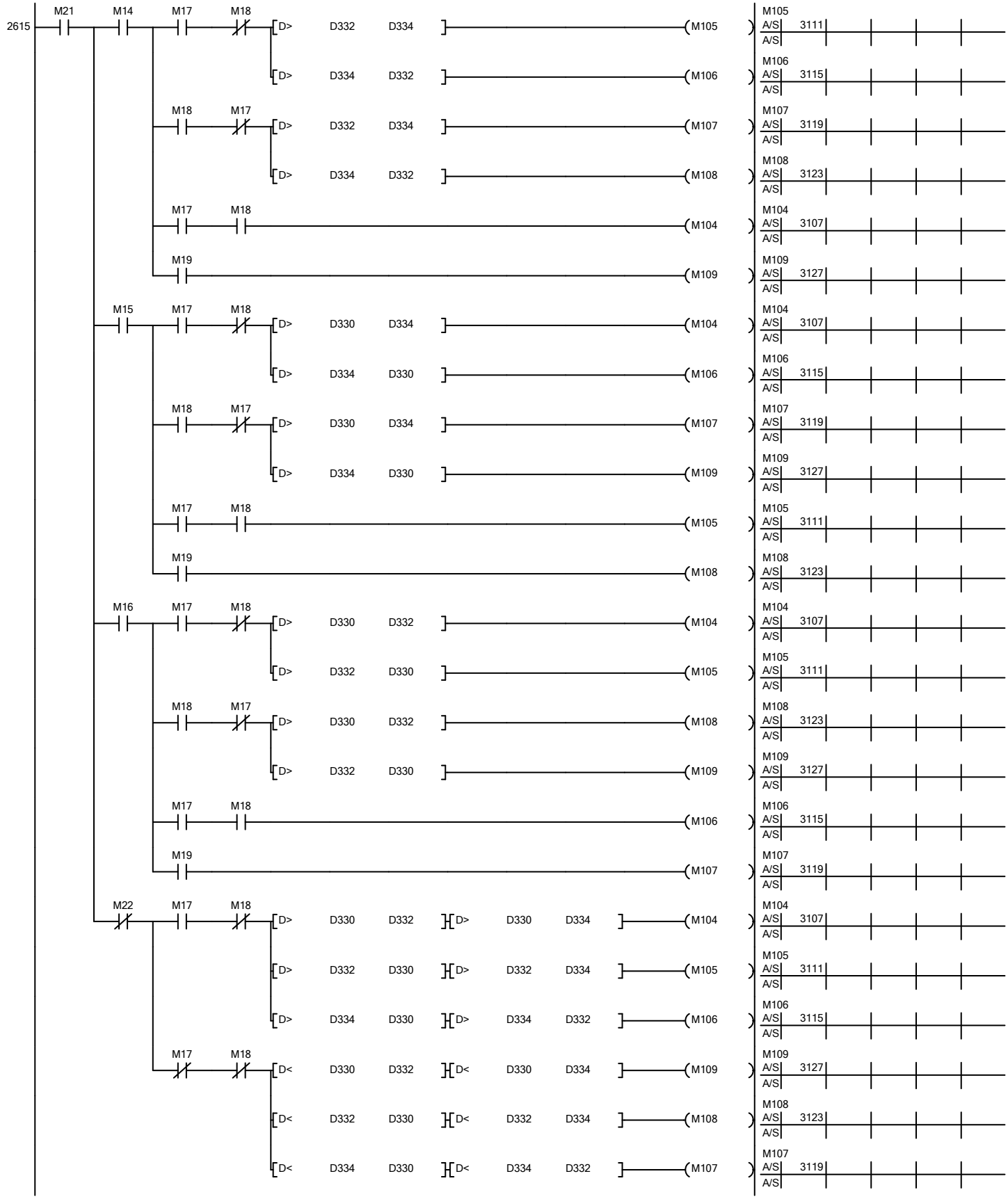

DEMUL D70 E0.4 D328
M702
DEADD D326 D328 D334
2287
M700
DEMOV E0 D330
2297
M701
DEMOV E0 D332
D326 A/S 2274 B/D 2247
D70 A/S 486 570 2260 3768 B/D 632
D328 A/S 2274 B/D 2260
D326 A/S 2274 B/D 2247
D328 A/S 2274 B/D 2260
D334 A/S 2461 2472 2493 2504 2565 B/D 2274 2308
D334 A/S 2585 2596 2605 2622 2633 B/D
D334 A/S 2647 2658 2681 2692 2706 B/D
D334 A/S 2717 2808 2828 2839 2848 B/D
D334 A/S 2871 2891 2902 2911 2937 B/D
D334 A/S 2948 2972 2983 3042 3062 B/D
D334 A/S 3073 3082 B/D
D330 A/S 2493 2504 2525 2536 2556 B/D 2191 2288
D330 A/S 2565 2576 2596 2681 2692 B/D
D330 A/S 2706 2717 2740 2751 2765 B/D
D330 A/S 2776 2799 2808 2819 2839 B/D
D330 A/S 2862 2871 2882 2902 2972 B/D
D330 A/S 2983 3007 3018 3033 3042 B/D
D330 A/S 3053 3073 B/D
D332 A/S 2461 2472 2525 2536 2556 B/D 2233 2298
D332 A/S 2576 2585 2605 2622 2633 B/D
D332 A/S 2647 2658 2740 2751 2765 B/D
D332 A/S 2776 2799 2819 2828 2848 B/D
D332 A/S 2862 2882 2891 2911 2937 B/D
D332 A/S 2948 3007 3018 3033 3053 B/D
D332 A/S 3062 3082 B/D

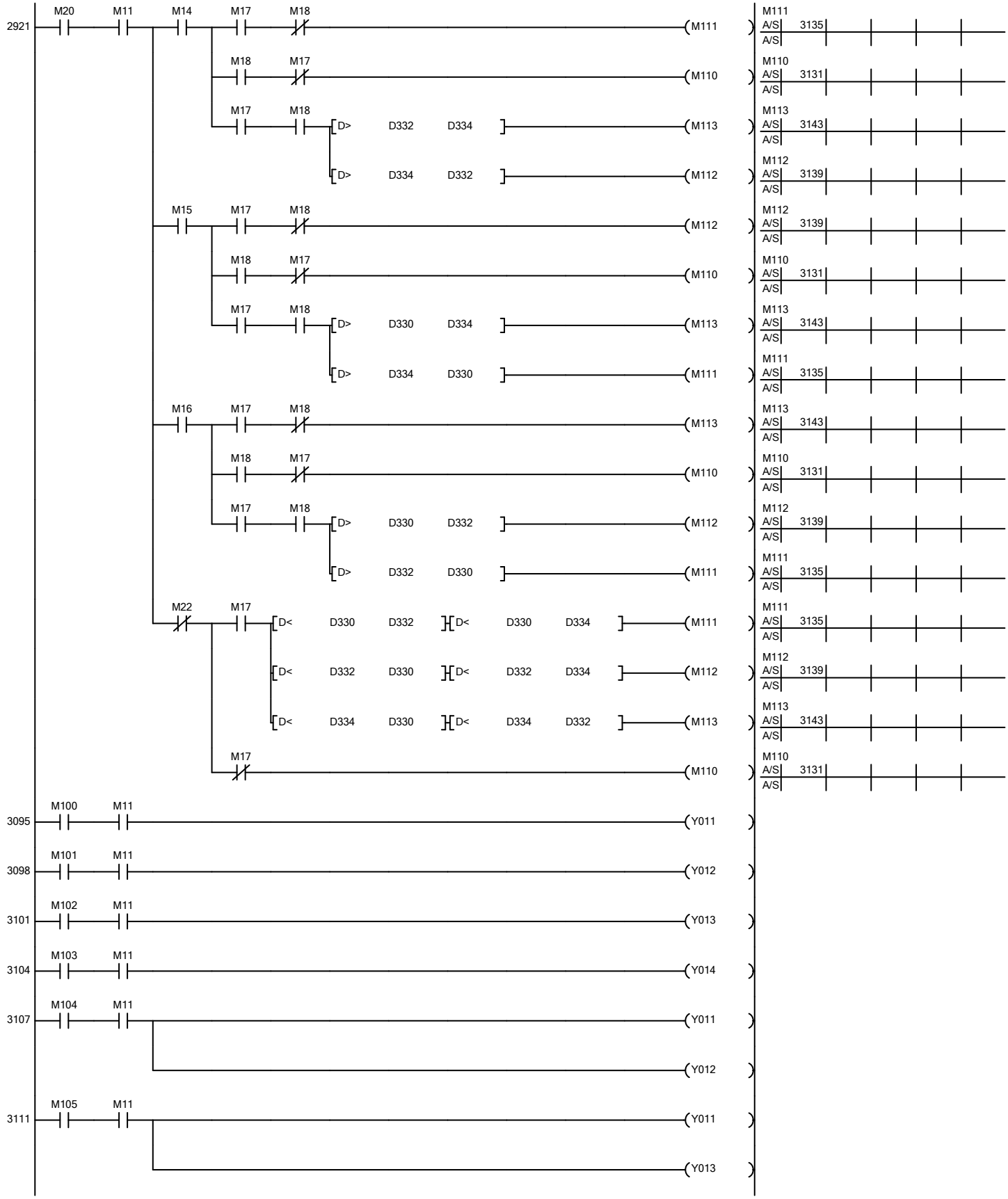


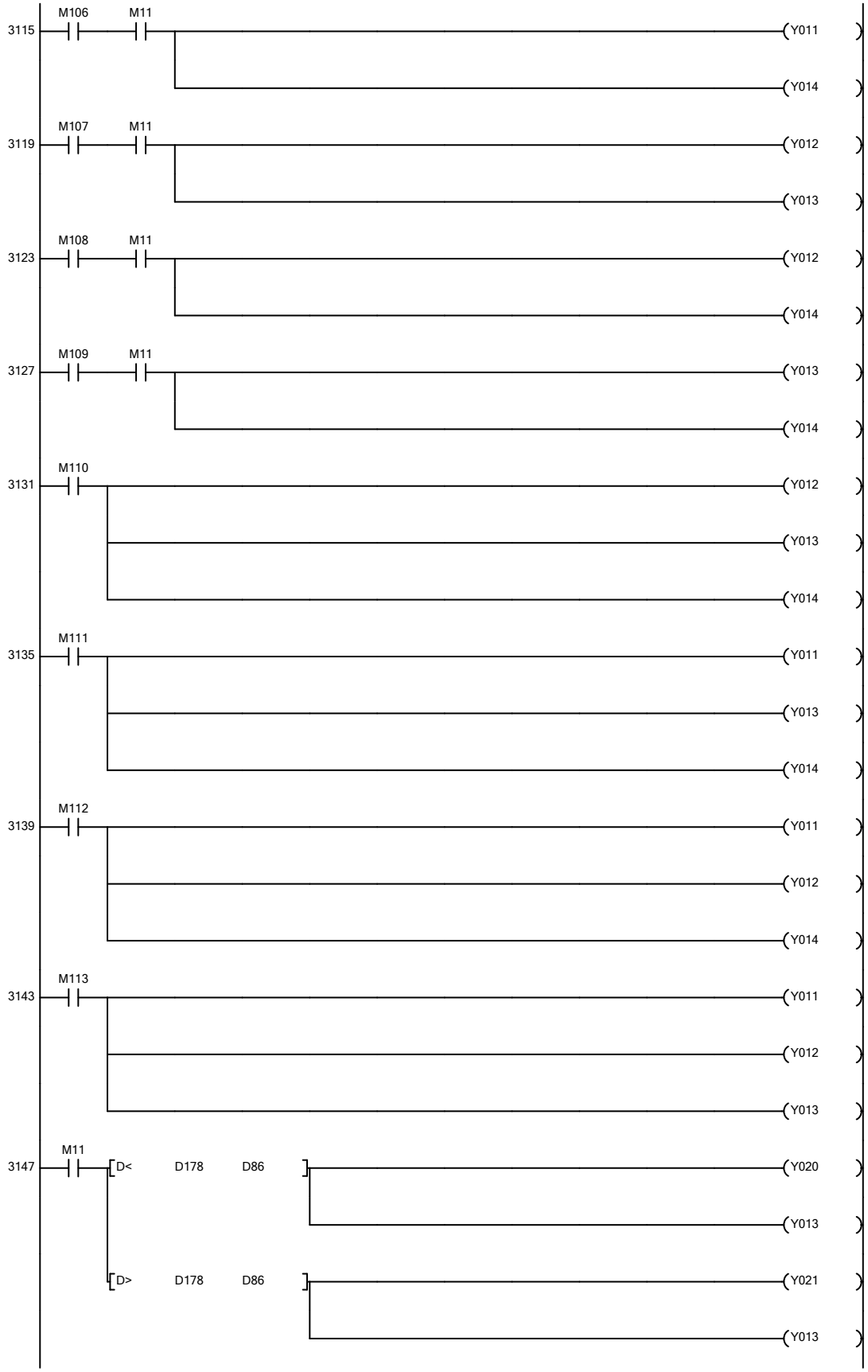
D334
A/S 2461 2472 2493 2504 2565
B/D 2274 2308
D334
A/S 2585 2596 2605 2622 2633
B/D
D334
A/S 2647 2658 2681 2692 2706
B/D
D334
A/S 2717 2808 2828 2839 2848
B/D
D334
A/S 2871 2891 2902 2911 2937
B/D
D334
A/S 2948 2972 2983 3042 3062
B/D
D334
A/S 3073 3082
B/D
M17
A/S 2371 2378 2390 2393 2409
B/D 2368 2380 2397 2412 2416
M17
A/S 2453 2485 2517 2619 2669
B/D 2645 2704 2763 2859 2931
M17
A/S 2678 2728 2737 2787 2796
B/D 2966 3001 3093
M17
A/S 2926 2934 2961 2969 2996
B/D
M17
A/S 3004 3031
B/D
M18
A/S 2372 2381 2391 2398 2413
B/D 2369 2379 2394 2410 2417
M18
A/S 2456 2488 2520 2644 2670
B/D 2620 2679 2738 2797 2860
M18
A/S 2703 2729 2762 2788 2930
B/D 2927 2962 2997
M18
A/S 2935 2965 2970 3000 3005
B/D
M19
A/S 2353 2373 2395 2399 2418
B/D 2392 2411 2414
M19
A/S 2459 2491 2523 2673 2732
B/D
M19
A/S 2791
B/D
M24
A/S 2377 2401
B/D 2375
M25
A/S 2374
B/D 2402
M20
A/S 2425 2549 2921
A/S

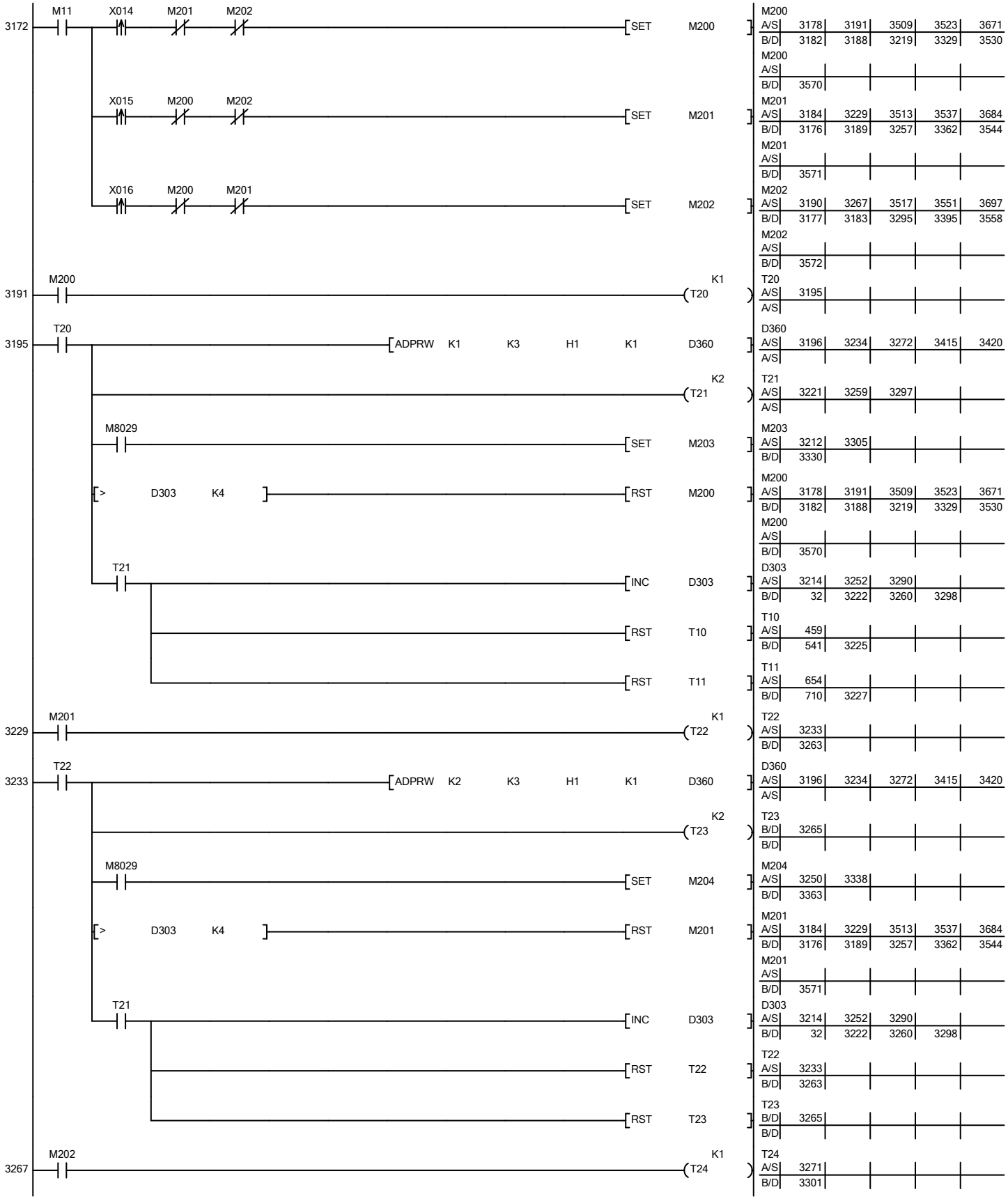

2390
M17 M18 M19 M11 Y003
M17 M18 M19 M21
M17 M18 M19
M24 M25
M21 A/S 2433 2615 A/S
2409
M17 M18 M19 M11 Y002
M17 M18 M19 M22
M17 M18 M19
M22 A/S 2441 2449 2552 B/D 2794 3029
2425
M20 $MOV "FSM" D480
D480 B/D 2426 2434 2442 B/D
2433
M21 $MOV "HSM" D480
D480 B/D 2426 2434 2442 B/D
2441
M22 $MOV "LSM" D480
D480 B/D 2426 2434 2442 B/D
2449
M22 M14 M17 M100
M100 A/S 3095 A/S
M18 M101
M101 A/S 3098 A/S
M19 D> D332 D334 M102
M102 A/S 3101 A/S
D> D334 D332 M103
M103 A/S 3104 A/S
M15 M17 M100
M100 A/S 3095 A/S
M18 M102
M102 A/S 3101 A/S
M19 D> D330 D334 M101
M101 A/S 3098 A/S
D> D334 D330 M103
M103 A/S 3104 A/S
M16 M17 M100
M100 A/S 3095 A/S
M18 M103
M103 A/S 3104 A/S
M19 D> D330 D332 M101
M101 A/S 3098 A/S
D> D332 D330 M102
M102 A/S 3101 A/S
M23 M20 M100
M100 A/S 3095 A/S
M22 M117
M117 A/S 2554 A/S

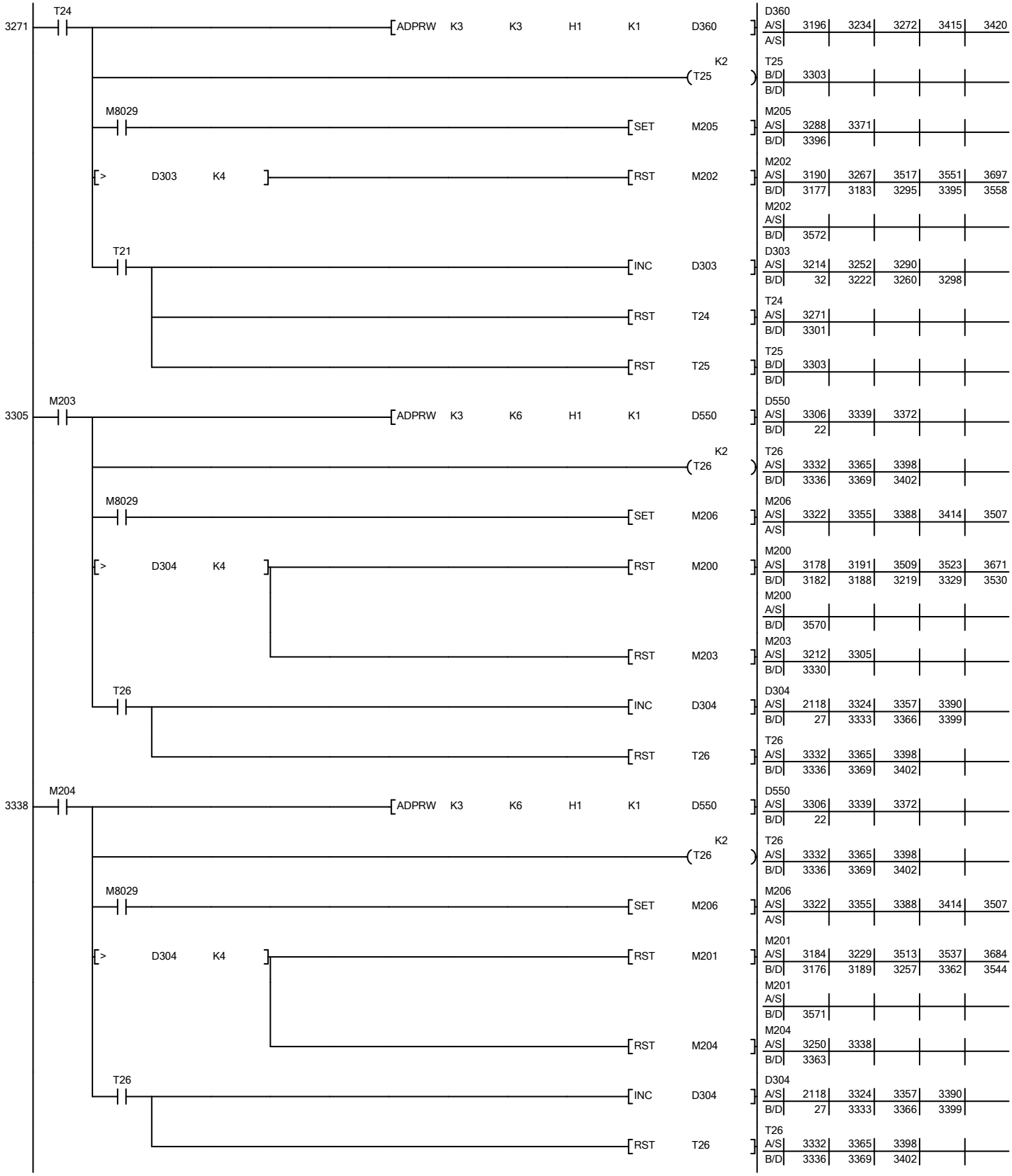

2554
M117
D>
D330
D332
D>
D330
D334
M101
D>
D332
D330
D>
D332
D334
M102
D>
D334
D330
D>
D334
D332
M103
M101
A/S
3098
A/S
M102
A/S
3101
A/S
M103
A/S
3104
A/S

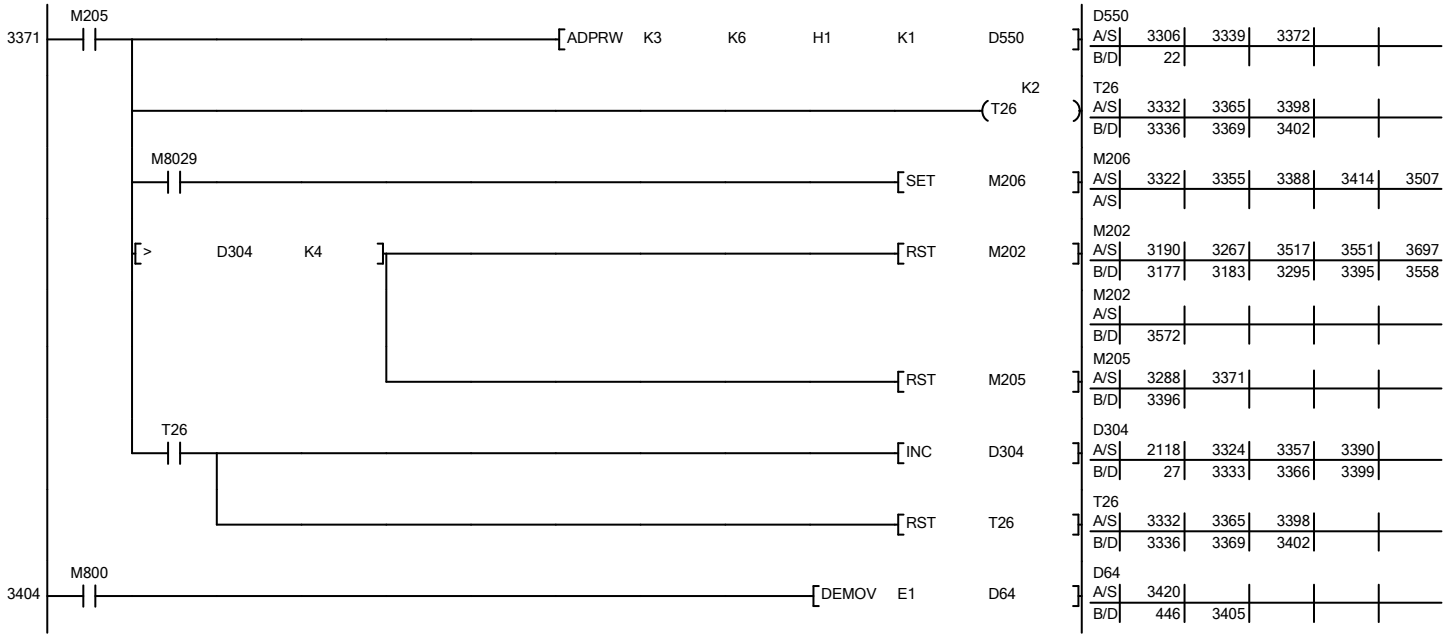

2615
M21 M14 M17 M18 D> D332 D334 M105
D> D334 D332 M106
M18 M17 D> D332 D334 M107
D> D334 D332 M108
M17 M18 M104
M19 M109
M15 M17 M18 D> D330 D334 M104
D> D334 D330 M106
M18 M17 D> D330 D334 M107
D> D334 D330 M109
M17 M18 M105
M19 M108
M16 M17 M18 D> D330 D332 M104
D> D332 D330 M105
M18 M17 D> D330 D332 M108
D> D332 D330 M109
M17 M18 M106
M19 M107
M22 M17 M18 D> D330 D332 D> D330 D334 M104
D> D332 D330 D> D332 D334 M105
D> D334 D330 D> D334 D332 M106
M17 M18 D< D330 D332 D< D330 D334 M109
D< D332 D330 D< D332 D334 M108
D< D334 D330 D< D334 D332 M107
M105 A/S 3111 A/S
M106 A/S 3115 A/S
M107 A/S 3119 A/S
M108 A/S 3123 A/S
M104 A/S 3107 A/S
M109 A/S 3127 A/S

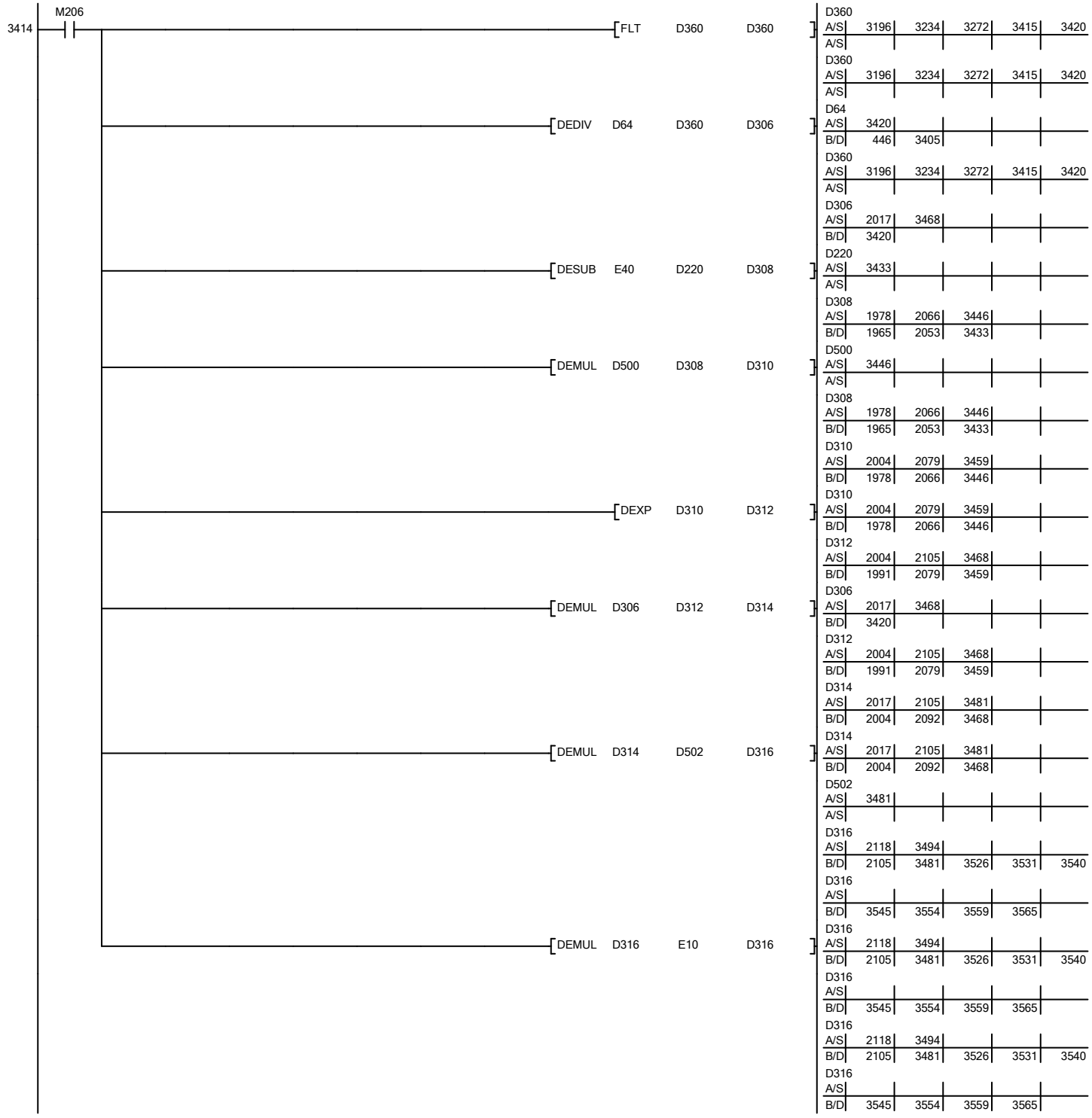

2921
M20 M11 M14 M17 M18 M111
M18 M17 M110
M17 M18 D> D332 D334 M113
D> D334 D332 M112
M15 M17 M18 M112
M18 M17 M110
M17 M18 D> D330 D334 M113
D> D334 D330 M111
M16 M17 M18 M113
M18 M17 M110
M17 M18 D> D330 D332 M112
D> D332 D330 M111
M22 M17 D< D330 D332][D< D330 D334 M111
D< D332 D330][D< D332 D334 M112
D< D334 D330][D< D334 D332 M113
M17 M110
3095
M100 M11 Y011
3098
M101 M11 Y012
3101
M102 M11 Y013
3104
M103 M11 Y014
3107
M104 M11 Y011
Y012
3111
M105 M11 Y011
Y013
M111 A/S 3135 A/S
M110 A/S 3131 A/S
M113 A/S 3143 A/S
M112 A/S 3139 A/S
M112 A/S 3139 A/S
M110 A/S 3131 A/S
M113 A/S 3143 A/S
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M112 A/S 3139 A/S
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M110 A/S 3131 A/S

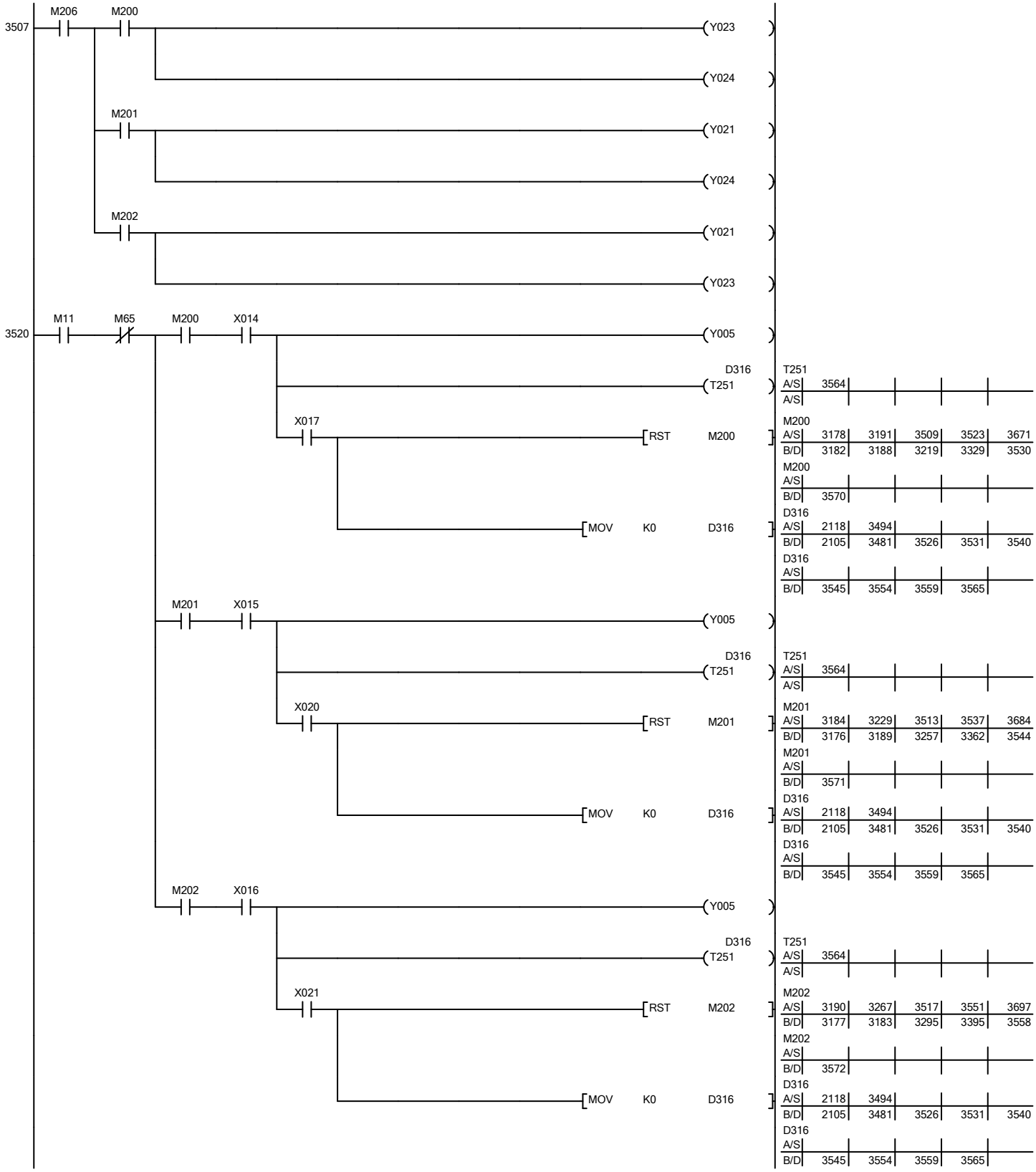

3115
M106
M11
Y011
Y014
3119
M107
M11
Y012
Y013
3123
M108
M11
Y012
Y014
3127
M109
M11
Y013
Y014
3131
M110
Y012
Y013
Y014
3135
M111
Y011
Y013
Y014
3139
M112
Y011
Y012
Y014
3143
M113
Y011
Y012
Y013
3147
M11
D< D178 D86
Y020
Y013
D> D178 D86
Y021
Y013

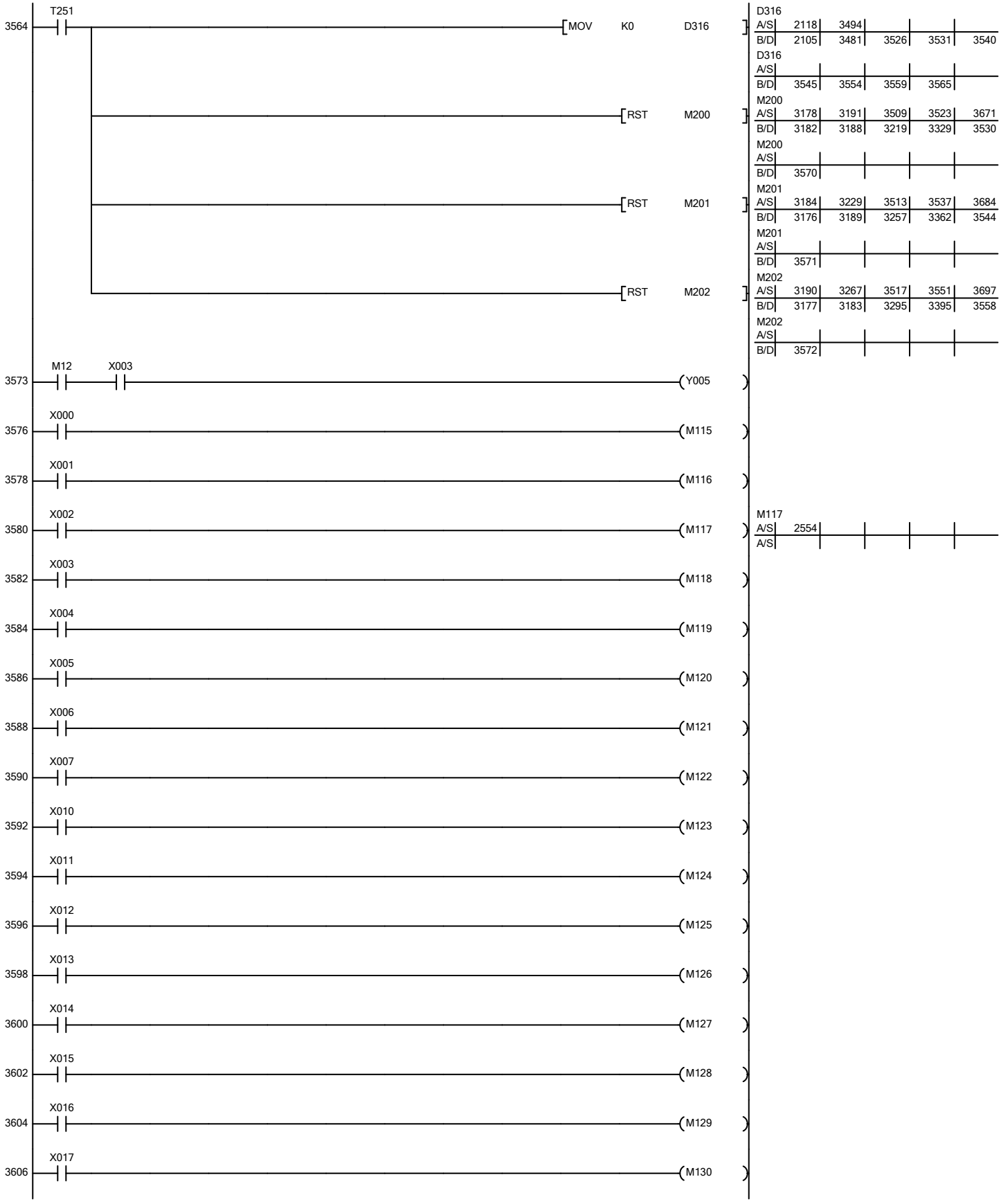

3172
M11
X014
M201
M202
SET M200
X015
M200
M202
SET M201
X016
M200
M201
SET M202
3191
M200
K1
T20
3195
T20
ADPRW K1 K3 H1 K1 D360
K2
T21
M8029
SET M203
> D303 K4
RST M200
T21
INC D303
RST T10
RST T11
3229
M201
K1
T22
3233
T22
ADPRW K2 K3 H1 K1 D360
K2
T23
M8029
SET M204
> D303 K4
RST M201
T21
INC D303
RST T22
RST T23
3267
M202
K1
T24
M200 A/S 3178 3191 3509 3523 3671
B/D 3182 3188 3219 3329 3530
M200 A/S B/D 3570
M201 A/S 3184 3229 3513 3537 3684
B/D 3176 3189 3257 3362 3544
M201 A/S B/D 3571
M202 A/S 3190 3267 3517 3551 3697
B/D 3177 3183 3295 3395 3558
M202 A/S B/D 3572
T20 A/S 3195 A/S
D360 A/S 3196 3234 3272 3415 3420 A/S
T21 A/S 3221 3259 3297 A/S
M203 A/S 3212 3305 B/D 3330
M200 A/S 3178 3191 3509 3523 3671
B/D 3182 3188 3219 3329 3530
M200 A/S B/D 3570
D303 A/S 3214 3252 3290 B/D 32 3222 3260 3298
T10 A/S 459 B/D 541 3225
T11 A/S 654 B/D 710 3227
T22 A/S 3233 B/D 3263
D360 A/S 3196 3234 3272 3415 3420 A/S
T23 B/D 3265 B/D
M204 A/S 3250 3338 B/D 3363
M201 A/S 3184 3229 3513 3537 3684
B/D 3176 3189 3257 3362 3544
M201 A/S B/D 3571
D303 A/S 3214 3252 3290 B/D 32 3222 3260 3298
T22 A/S 3233 B/D 3263
T23 B/D 3265 B/D
T24 A/S 3271 B/D 3301

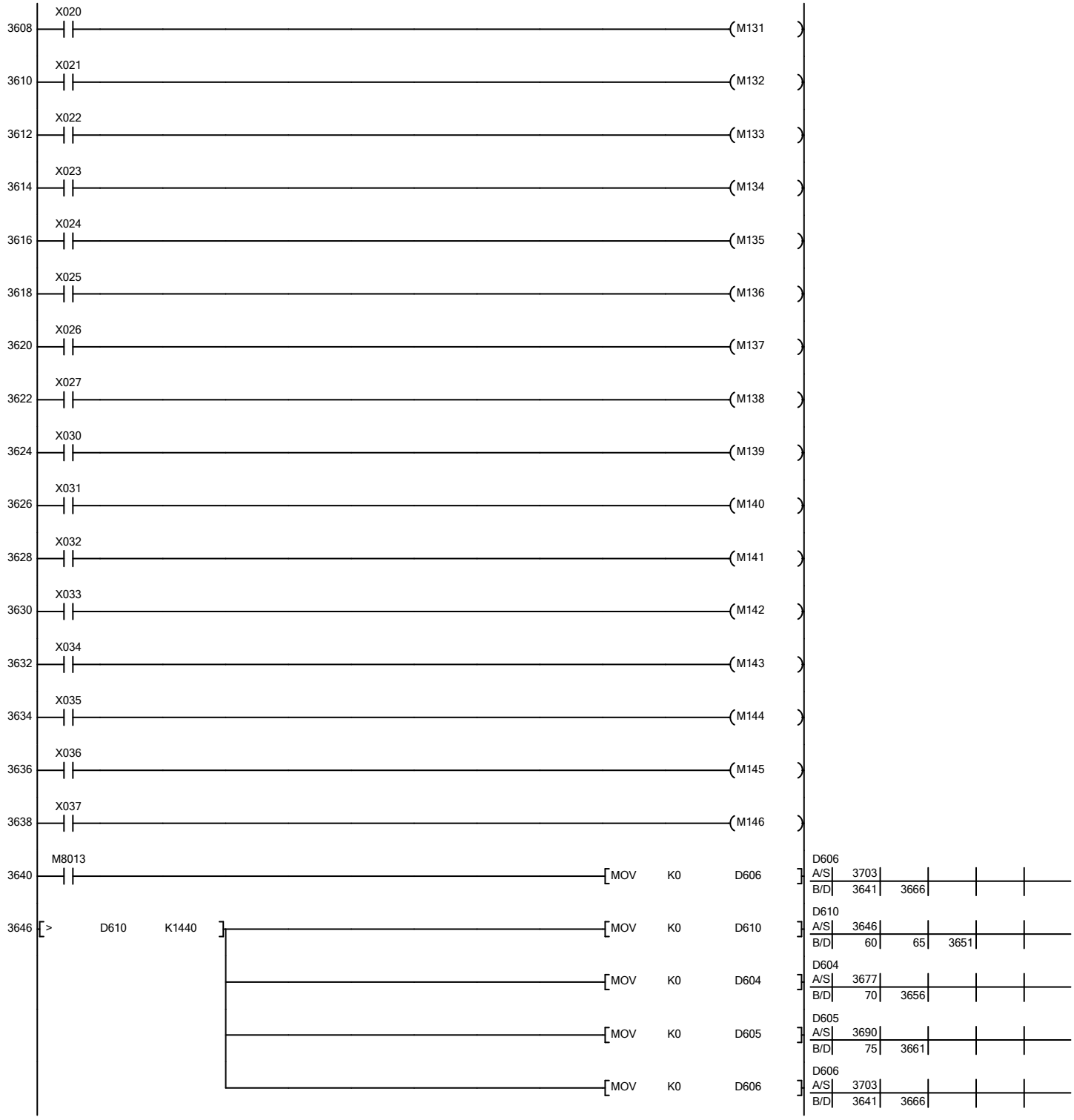

3271
T24
ADPRW K3 K3 H1 K1 D360
K2
T25
M8029
SET M205
> D303 K4
RST M202
T21
INC D303
RST T24
RST T25
3305
M203
ADPRW K3 K6 H1 K1 D550
K2
T26
M8029
SET M206
> D304 K4
RST M200
RST M203
T26
INC D304
RST T26
3338
M204
ADPRW K3 K6 H1 K1 D550
K2
T26
M8029
SET M206
> D304 K4
RST M201
RST M204
T26
INC D304
RST T26
D360 A/S 3196 3234 3272 3415 3420 A/S
T25 B/D 3303 B/D
M205 A/S 3288 3371 B/D 3396
M202 A/S 3190 3267 3517 3551 3697 B/D 3177 3183 3295 3395 3558
M202 A/S B/D 3572
D303 A/S 3214 3252 3290 B/D 32 3222 3260 3298
T24 A/S 3271 B/D 3301
T25 B/D 3303 B/D
D550 A/S 3306 3339 3372 B/D 22
T26 A/S 3332 3365 3398 B/D 3336 3369 3402
M206 A/S 3322 3355 3388 3414 3507 A/S
M200 A/S 3178 3191 3509 3523 3671 B/D 3182 3188 3219 3329 3530
M200 A/S B/D 3570
M203 A/S 3212 3305 B/D 3330
D304 A/S 2118 3324 3357 3390 B/D 27 3333 3366 3399
T26 A/S 3332 3365 3398 B/D 3336 3369 3402
D550 A/S 3306 3339 3372 B/D 22
T26 A/S 3332 3365 3398 B/D 3336 3369 3402
M206 A/S 3322 3355 3388 3414 3507 A/S
M201 A/S 3184 3229 3513 3537 3684 B/D 3176 3189 3257 3362 3544
M201 A/S B/D 3571
M204 A/S 3250 3338 B/D 3363
D304 A/S 2118 3324 3357 3390 B/D 27 3333 3366 3399
T26 A/S 3332 3365 3398 B/D 3336 3369 3402

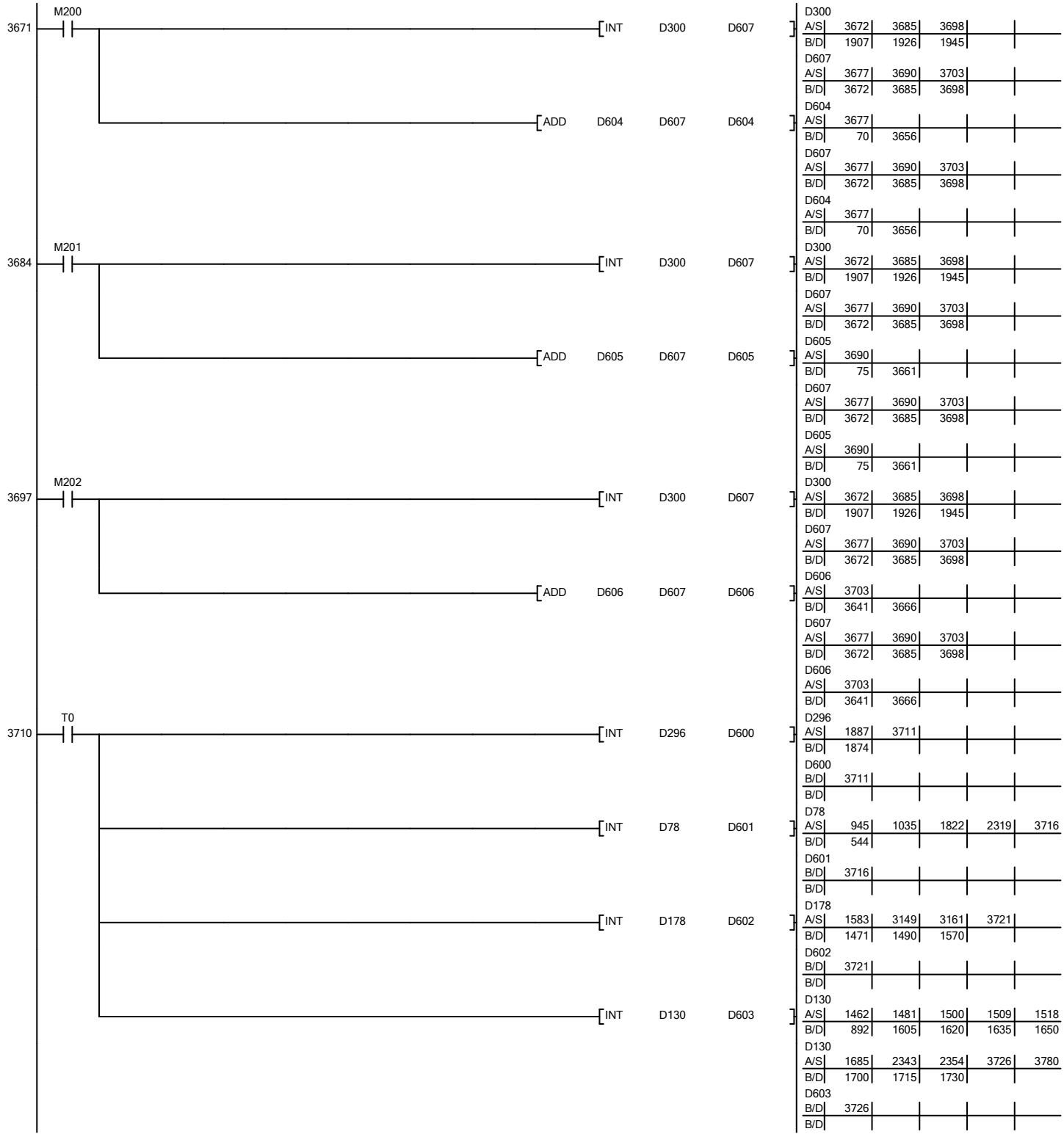

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3371
ADPRW K3 K6 H1 K1 D550
K2
T26
M8029
SET M206
> D304 K4
RST M202
RST M205
T26
INC D304
RST T26
M800
3404
DEMOV E1 D64
D550
A/S 3306 3339 3372
B/D 22
T26
A/S 3332 3365 3398
B/D 3336 3369 3402
M206
A/S 3322 3355 3388 3414 3507
A/S
M202
A/S 3190 3267 3517 3551 3697
B/D 3177 3183 3295 3395 3558
M202
A/S
B/D 3572
M205
A/S 3288 3371
B/D 3396
D304
A/S 2118 3324 3357 3390
B/D 27 3333 3366 3399
T26
A/S 3332 3365 3398
B/D 3336 3369 3402
D64
A/S 3420
B/D 446 3405

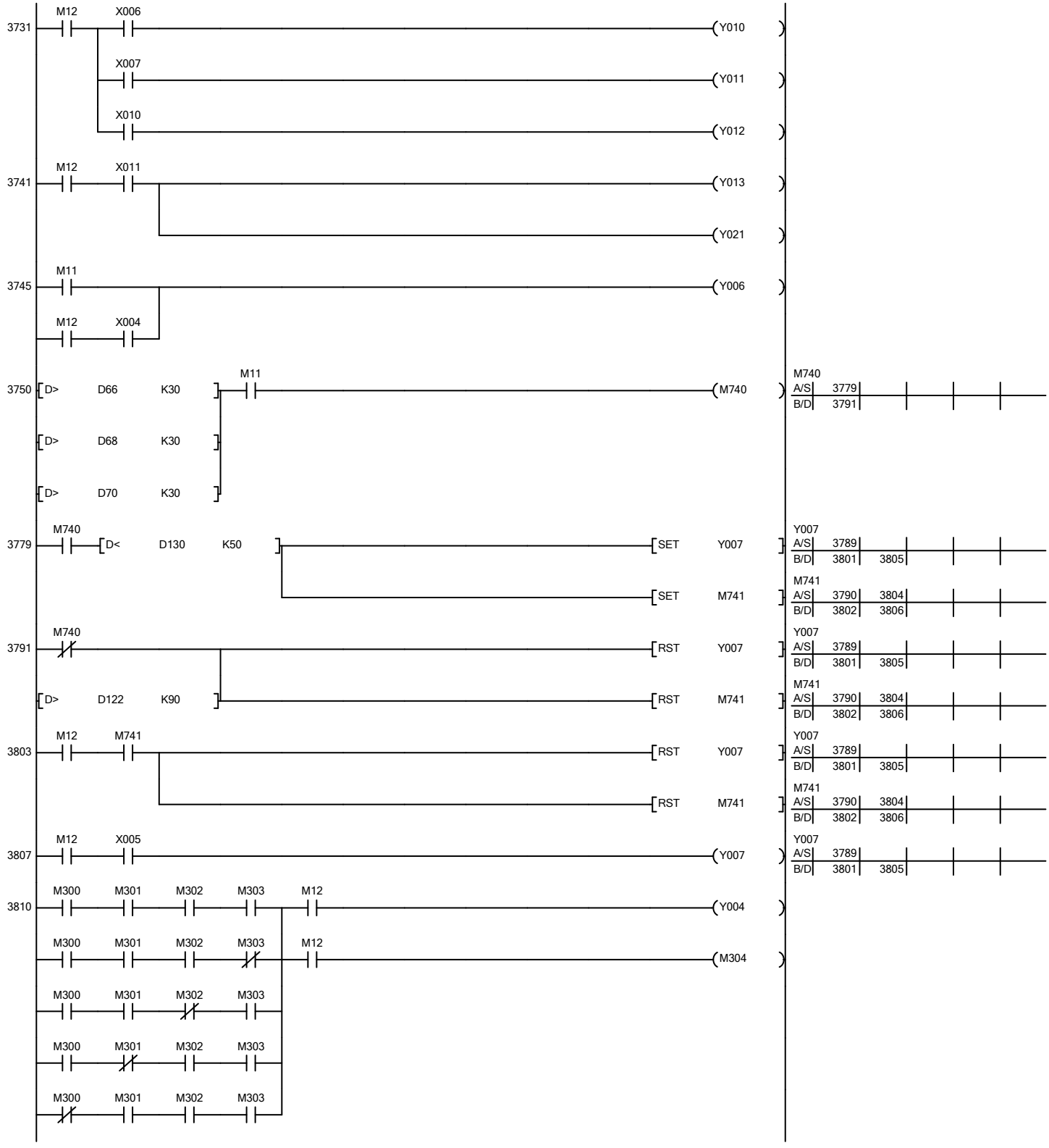

M206
3414
FLT D360 D360
DEDIV D64 D360 D306
DESUB E40 D220 D308
DEMUL D500 D308 D310
DEXP D310 D312
DEMUL D306 D312 D314
DEMUL D314 D502 D316
DEMUL D316 E10 D316
D360 A/S 3196 3234 3272 3415 3420
D64 A/S 3420 B/D 446 3405
D306 A/S 2017 3468 B/D 3420
D220 A/S 3433
D308 A/S 1978 2066 3446 B/D 1965 2053 3433
D500 A/S 3446
D310 A/S 2004 2079 3459 B/D 1978 2066 3446
D312 A/S 2004 2105 3468 B/D 1991 2079 3459
D314 A/S 2017 2105 3481 B/D 2004 2092 3468
D502 A/S 3481
D316 A/S 2118 3494 B/D 2105 3481 3526 3531 3540
B/D 3545 3554 3559 3565

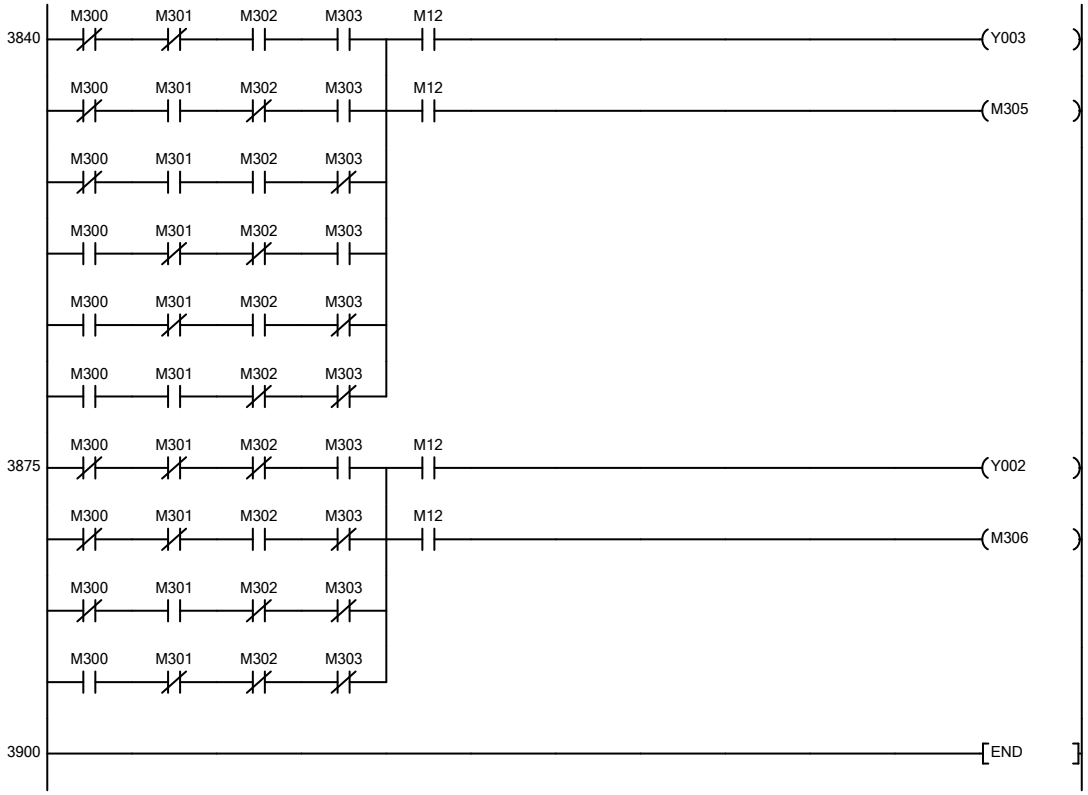

3507
M206
M200
Y023
Y024
M201
Y021
Y024
M202
Y021
Y023
3520
M11
M65
M200
X014
Y005
D316
T251
A/S 3564
X017
RST M200
M200
A/S 3178 3191 3509 3523 3671
B/D 3182 3188 3219 3329 3530
M200
A/S
B/D 3570
MOV K0 D316
D316
A/S 2118 3494
B/D 2105 3481 3526 3531 3540
D316
A/S
B/D 3545 3554 3559 3565
M201
X015
Y005
D316
T251
A/S 3564
X020
RST M201
M201
A/S 3184 3229 3513 3537 3684
B/D 3176 3189 3257 3362 3544
M201
A/S
B/D 3571
MOV K0 D316
D316
A/S 2118 3494
B/D 2105 3481 3526 3531 3540
D316
A/S
B/D 3545 3554 3559 3565
M202
X016
Y005
D316
T251
A/S 3564
X021
RST M202
M202
A/S 3190 3267 3517 3551 3697
B/D 3177 3183 3295 3395 3558
M202
A/S
B/D 3572
MOV K0 D316
D316
A/S 2118 3494
B/D 2105 3481 3526 3531 3540
D316
A/S
B/D 3545 3554 3559 3565


3564 T251 MOV K0 D316
RST M200
RST M201
RST M202
3573 M12 X003 Y005
3576 X000 M115
3578 X001 M116
3580 X002 M117
3582 X003 M118
3584 X004 M119
3586 X005 M120
3588 X006 M121
3590 X007 M122
3592 X010 M123
3594 X011 M124
3596 X012 M125
3598 X013 M126
3600 X014 M127
3602 X015 M128
3604 X016 M129
3606 X017 M130
D316 A/S 2118 3494 B/D 2105 3481 3526 3531 3540
D316 A/S B/D 3545 3554 3559 3565
M200 A/S 3178 3191 3509 3523 3671 B/D 3182 3188 3219 3329 3530
M200 A/S B/D 3570
M201 A/S 3184 3229 3513 3537 3684 B/D 3176 3189 3257 3362 3544
M201 A/S B/D 3571
M202 A/S 3190 3267 3517 3551 3697 B/D 3177 3183 3295 3395 3558
M202 A/S B/D 3572
M117 A/S 2554 A/S


3608 X020 M131
3610 X021 M132
3612 X022 M133
3614 X023 M134
3616 X024 M135
3618 X025 M136
3620 X026 M137
3622 X027 M138
3624 X030 M139
3626 X031 M140
3628 X032 M141
3630 X033 M142
3632 X034 M143
3634 X035 M144
3636 X036 M145
3638 X037 M146
3640 M8013 MOV K0 D606
3646 > D610 K1440
MOV K0 D610
MOV K0 D604
MOV K0 D605
MOV K0 D606
D606 A/S 3703 B/D 3641 3666
D610 A/S 3646 B/D 60 65 3651
D604 A/S 3677 B/D 70 3656
D605 A/S 3690 B/D 75 3661
D606 A/S 3703 B/D 3641 3666


M200
3671
INT D300 D607
ADD D604 D607 D604
M201
3684
INT D300 D607
ADD D605 D607 D605
M202
3697
INT D300 D607
ADD D606 D607 D606
T0
3710
INT D296 D600
INT D78 D601
INT D178 D602
INT D130 D603
D300 A/S 3672 3685 3698 B/D 1907 1926 1945
D607 A/S 3677 3690 3703 B/D 3672 3685 3698
D604 A/S 3677 B/D 70 3656
D607 A/S 3677 3690 3703 B/D 3672 3685 3698
D604 A/S 3677 B/D 70 3656
D300 A/S 3672 3685 3698 B/D 1907 1926 1945
D607 A/S 3677 3690 3703 B/D 3672 3685 3698
D605 A/S 3690 B/D 75 3661
D607 A/S 3677 3690 3703 B/D 3672 3685 3698
D605 A/S 3690 B/D 75 3661
D300 A/S 3672 3685 3698 B/D 1907 1926 1945
D607 A/S 3677 3690 3703 B/D 3672 3685 3698
D606 A/S 3703 B/D 3641 3666
D607 A/S 3677 3690 3703 B/D 3672 3685 3698
D606 A/S 3703 B/D 3641 3666
D296 A/S 1887 3711 B/D 1874
D600 B/D 3711 B/D
D78 A/S 945 1035 1822 2319 3716 B/D 544
D601 B/D 3716 B/D
D178 A/S 1583 3149 3161 3721 B/D 1471 1490 1570
D602 B/D 3721 B/D
D130 A/S 1462 1481 1500 1509 1518 B/D 892 1605 1620 1635 1650
D130 A/S 1685 2343 2354 3726 3780 B/D 1700 1715 1730
D603 B/D 3726 B/D


3731 M12 X006 Y010
X007 Y011
X010 Y012
3741 M12 X011 Y013
Y021
3745 M11 Y006
M12 X004
3750 D> D66 K30 M11 M740
D> D68 K30
D> D70 K30
3779 M740 D< D130 K50 SET Y007
SET M741
3791 M740 RST Y007
D> D122 K90 RST M741
3803 M12 M741 RST Y007
RST M741
3807 M12 X005 Y007
3810 M300 M301 M302 M303 M12 Y004
M300 M301 M302 M303 M12 M304
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303
M740 A/S 3779 B/D 3791
Y007 A/S 3789 B/D 3801 3805
M741 A/S 3790 3804 B/D 3802 3806
Y007 A/S 3789 B/D 3801 3805
M741 A/S 3790 3804 B/D 3802 3806
Y007 A/S 3789 B/D 3801 3805
M741 A/S 3790 3804 B/D 3802 3806
Y007 A/S 3789 B/D 3801 3805


3840
M300 M301 M302 M303 M12
Y003
M300 M301 M302 M303 M12
M305
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303
3875
M300 M301 M302 M303 M12
Y002
M300 M301 M302 M303 M12
M306
M300 M301 M302 M303
M300 M301 M302 M303
3900
END

[Line Statement List]

Step	Line Statement
0	Timer
16	Declarations
80	Analog to Digital
114	ADC to Float
170	Temperature
327	Distance
432	Flowmeter
459	Temperature Averaging
654	Level Averaging
944	5 - sec Sensor Logging
1167	Error Threshold
1438	E-Stops
1451	Automation
1460	Water Temperature logging
1593	Heater latching
1768	Height to Mass Conversion
1821	Dynamic Upper Thermal Boundary
1897	PO Desingation
1954	PO Timer
2144	PO Dispensing
2162	Tank scoring
2317	Need Scheduler
2371	Mode Scheduler
2425	Mode Indicator
2449	LSM Scheduler
2615	HSM Scheduler
2921	FSM Scheduler
3095	Scheduler Gates
3147	Solar Gate
3172	Oil Scheduler
3404	Oil Calculation
3507	Oil Distribution Gating
3520	Oil Distribution Timing
3576	HMI Display for Inputs
3640	Oil Chart
3710	Charting
3731	Manual Heater Control
3741	Manual Solar Gate
3745	Jacket pump (Auto/manual)
3750	AODD (Auto/Manual)
3810	Manual Pump Mode